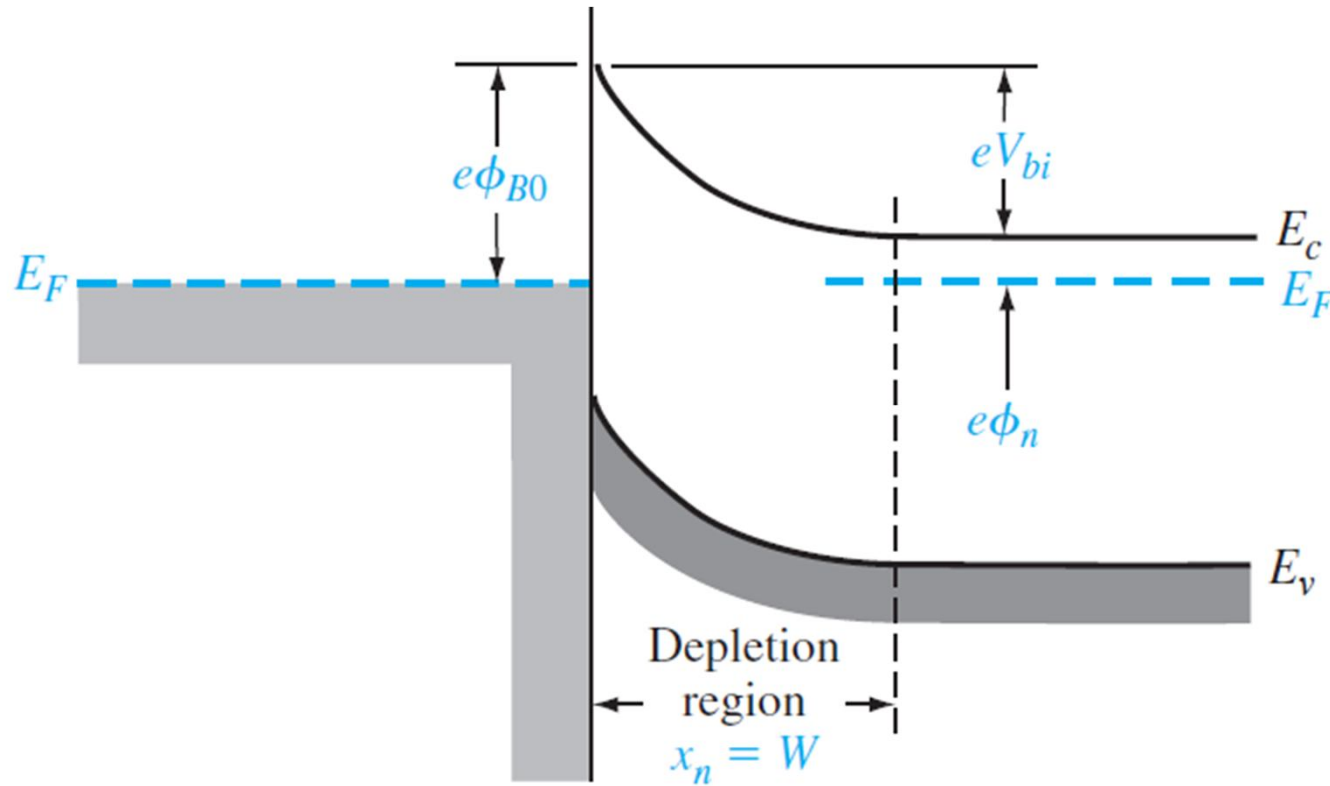


L9: Metal–Semiconductor and Semiconductor Heterojunctions



Dr. Shuming Chen

Department of Electrical and Electronic Engineering, SUSTech

Topics

- Determine the **energy - band diagram** of a metal–semiconductor junction.
- Investigate the electrostatics of the rectifying metal–semiconductor junction, which is known as the **Schottky barrier diode**.
- Derive the **ideal current–voltage** relation of the Schottky barrier diode.
- Discuss differences in the current transport mechanism between the Schottky barrier diode and pn junction diode, and discuss differences in turn - on voltage and switching times.
- Discuss **ohmic contacts**, which are low - resistance, nonrectifying metal–semiconductor junctions.
- Investigate the characteristics of a semiconductor **heterojunction**.

9.1 | THE SCHOTTKY BARRIER DIODE

- The parameter ϕ_m is the metal work function (measured in volts),
- ϕ_s is the semiconductor work function,
- χ is known as the electron affinity.

Table 9.1 | Work functions of some elements

Element	Work function, ϕ_m
Ag, silver	4.26
Al, aluminum	4.28
Au, gold	5.1
Cr, chromium	4.5
Mo, molybdenum	4.6
Ni, nickel	5.15
Pd, palladium	5.12
Pt, platinum	5.65
Ti, titanium	4.33
W, tungsten	4.55

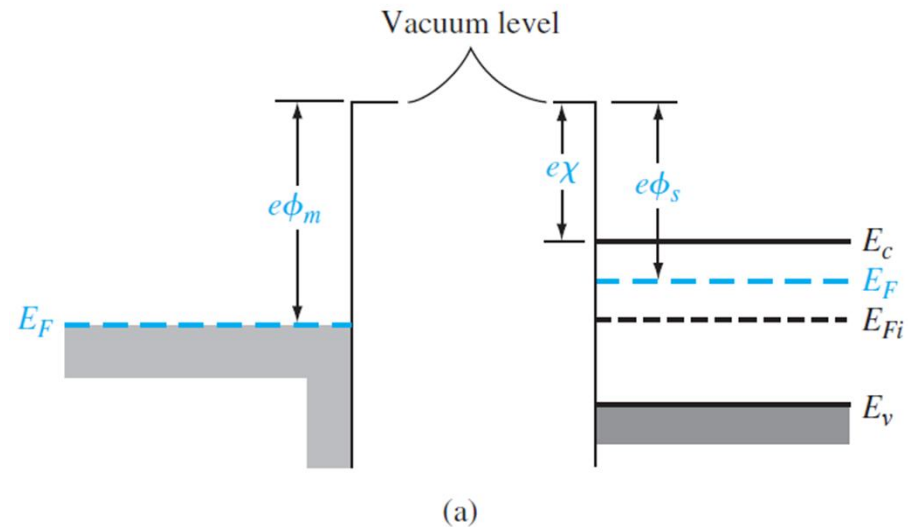


Table 9.2 | Electron affinity of some semiconductors

Element	Electron affinity, χ
Ge, germanium	4.13
Si, silicon	4.01
GaAs, gallium arsenide	4.07
AlAs, aluminum arsenide	3.5

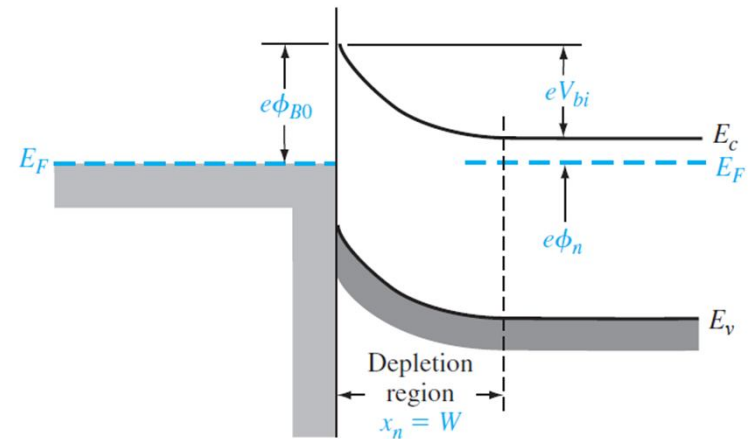
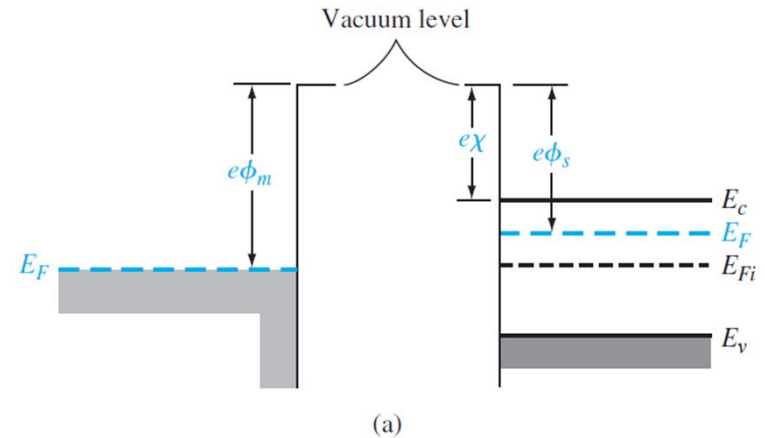
9.1 | THE SCHOTTKY BARRIER DIODE

- In order for the Fermi level to become a constant through the system in thermal equilibrium, electrons from the semiconductor flow into the lower energy states in the metal.
- Positively charged donor atoms remain in the semiconductor, creating a space charge region.
- This barrier is known as the *Schottky barrier* and is given, ideally, by:

$$\phi_{B0} = (\phi_m - \chi)$$

- On the semiconductor side, V_{bi} is the built-in potential barrier. This barrier, similar to the case of the pn junction, is the barrier seen by electrons in the conduction band trying to move into the metal. The built-in potential barrier is given by:

$$V_{bi} = \phi_{B0} - \phi_n$$



Schottky diode under bias

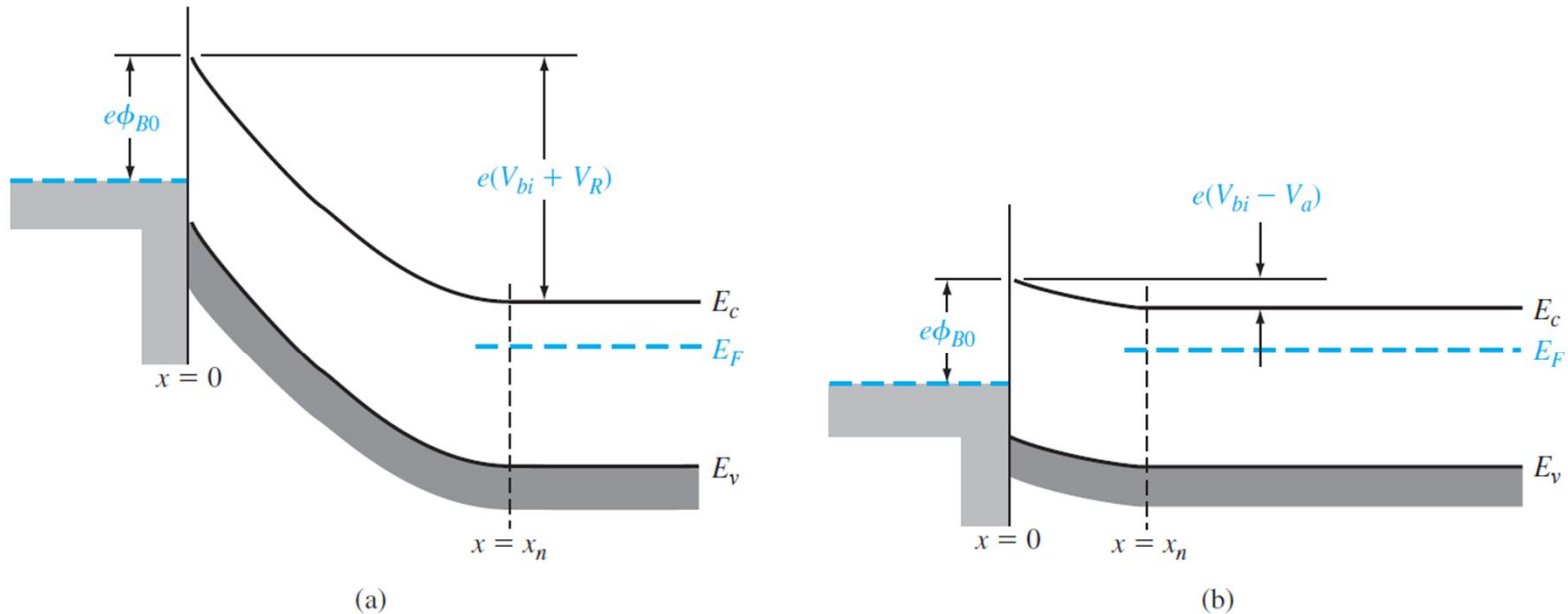
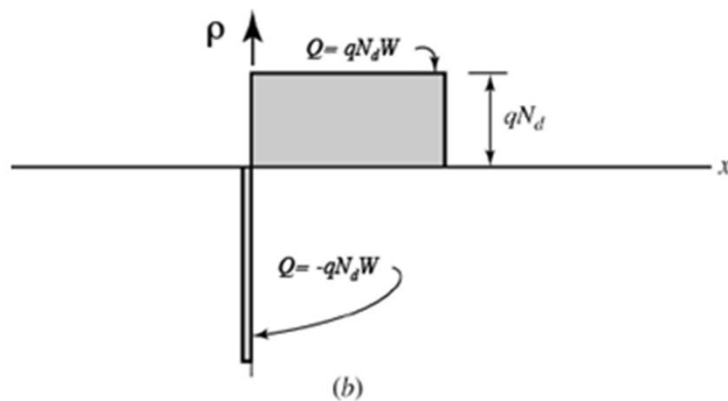
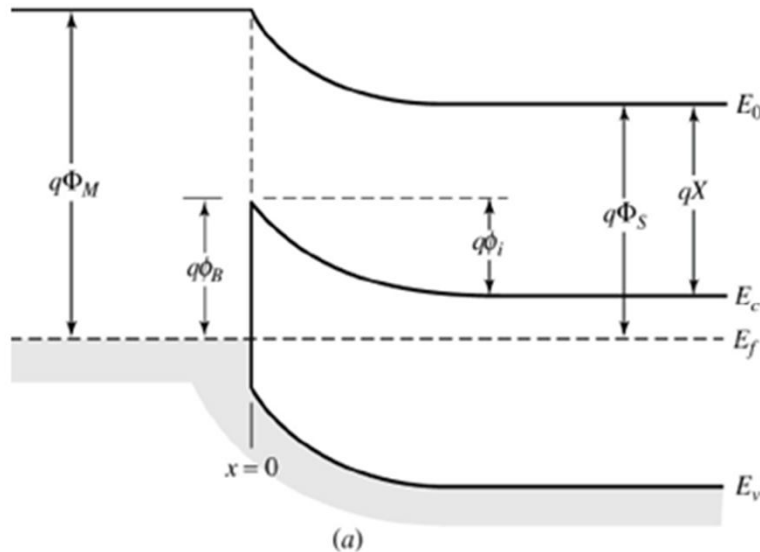


Figure 9.2 | Ideal energy-band diagram of a metal–semiconductor junction (a) under reverse bias and (b) under forward bias.

- V_R is the magnitude of the reverse - biased voltage and V_a is the magnitude of the forward - bias voltage.
- We expect the current–voltage characteristics of the Schottky barrier junction to be similar to the exponential behavior of the pn junction diode.
- The current mechanism here, however, is due to the flow of **majority** carrier electrons.

Ideal Junction Properties



To proceed, the following assumptions (depletion approximation) are made:

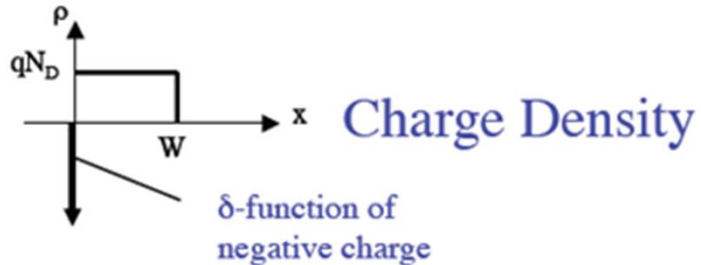
- 1) Neglect the free hole concentration for n-type material (opposite for p-type).
- 2) $n \approx 0$ in a region from $x = 0$ to $x = W$.
- 3) Beyond $x = W$, $n = N_D$.
- 4) $Q = q N_D W A =$ total charge (ionized dopants) in the depletion region.

The voltage across the depletion region is governed by Poisson's equation (assume complete ionization)

$$\frac{d^2V}{dx^2} = -\frac{dE}{dx} = \frac{q}{K_s \epsilon_o} \left[(\cancel{p} - \cancel{n}) - (N_D - \cancel{N}_A) \right]$$

$$\approx -\frac{q}{K_s \epsilon_o} N_D$$

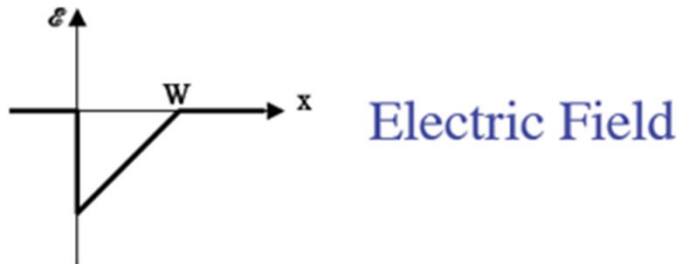
Ideal Junction Properties



Integrating once from an arbitrary point x in the depletion region to a point $x = W$,

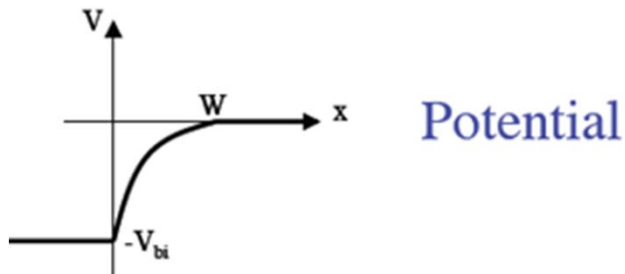
$$E(x) = -\frac{qN_D}{\kappa\epsilon_o}(W - x)$$

The maximum E field is at $x = 0$



$$E_{\max} = -\frac{qN_D}{K_s\epsilon_o}W$$

Integrating again,



$$V(x) = -\frac{qN_D}{2K_s\epsilon_o}(W - x)^2$$

Ideal Junction Properties

The total potential difference between metal and the semiconductor also known as the built-in voltage, V_{bi} , is

$$V_{bi} = q(\Phi_M - \Phi_S) = V(0) = \frac{1}{2} \frac{qN_D}{K_s \epsilon_o} W^2$$

The width of the depletion layer, W , is (at zero applied bias),

$$W = \sqrt{\frac{2K_s \epsilon_o V_{bi}}{qN_D}}$$

The charge in the depletion layer, Q , is

$$Q = qAWN_D = A\sqrt{2q\epsilon_o K_s N_D V_{bi}}$$

where A = cross-sectional area

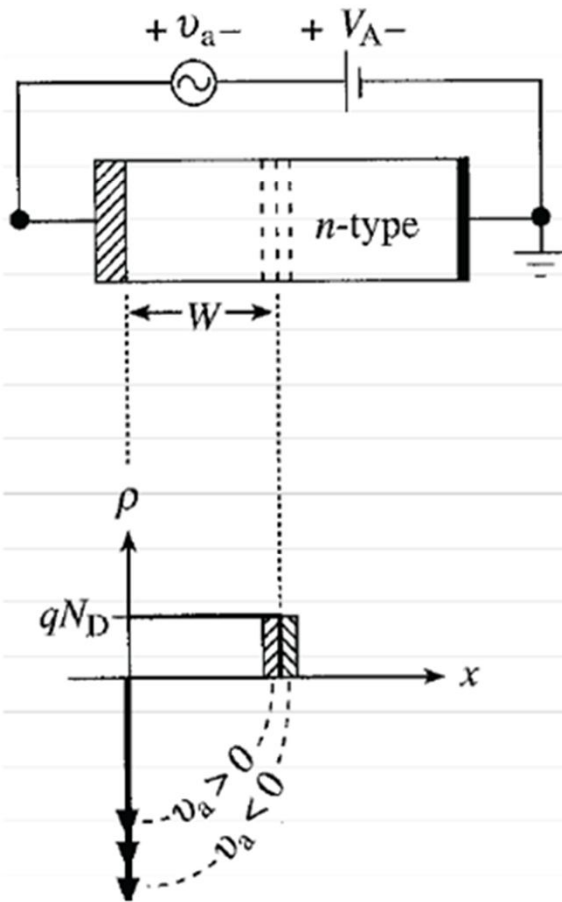
Capacitance

When a voltage of V_A is applied, in the depletion region,

$$Q = A\sqrt{2qK_s\epsilon_oN_D(V_{bi} - V_A)}$$

$$C = \frac{dQ}{dV} = \frac{K_s\epsilon_oA}{\left[\frac{2K_s\epsilon_o}{qN_D}(V_{bi} - V_A)\right]^{1/2}}$$

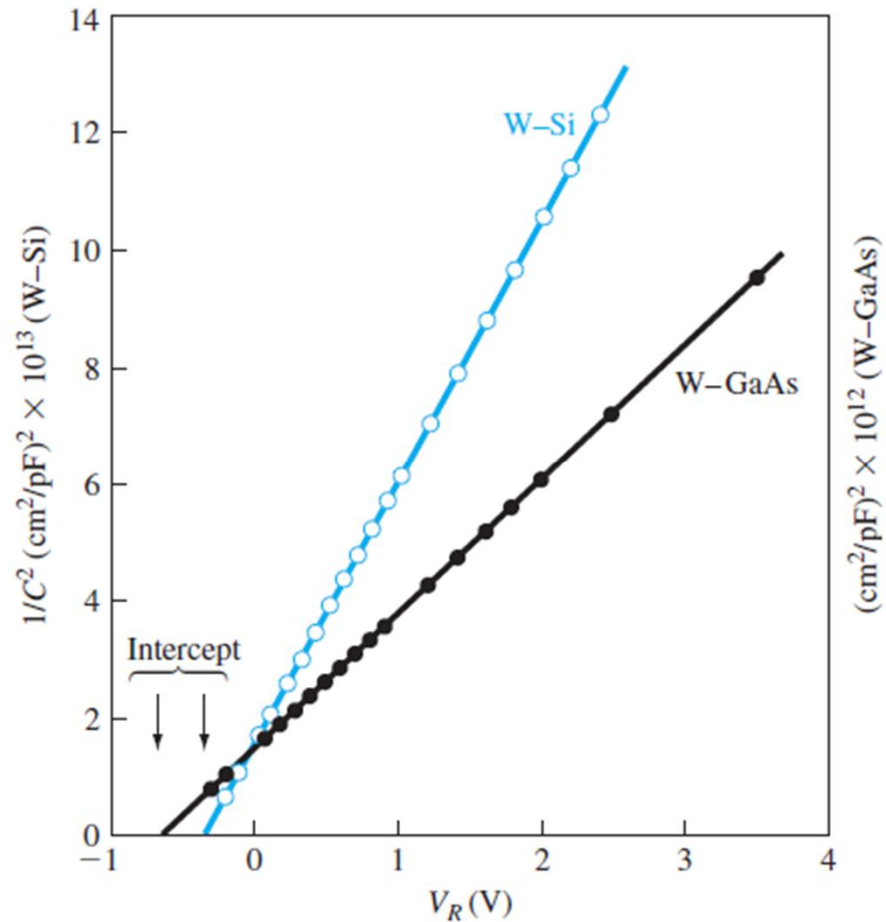
$$\frac{1}{C^2} = \frac{2}{qN_DK_s\epsilon_oA^2}(V_{bi} - V_A)$$



Capacitance

$$\left(\frac{1}{C'}\right)^2 = \frac{2(V_{bi} + V_R)}{e\epsilon_s N_d}$$

From the intercept of the curve, we can get the built-in potential barrier V_{bi} , and from the slope of the curve, we can get the semiconductor doping N_d .



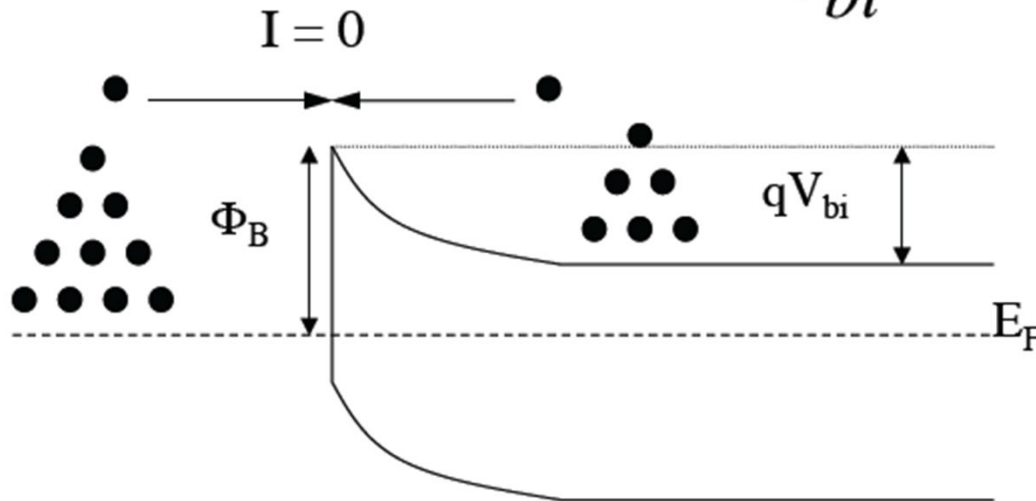
the built - in potential barrier of the gallium arsenide (with a smaller N_c , and thus to achieve similar n , E_F is more close to E_c , leading to a smaller ϕ_s) Schottky diode is larger than that of the silicon diode.

Schottky barrier diode

Equilibrium

$$\Phi_B = \Phi_M - \chi_S$$

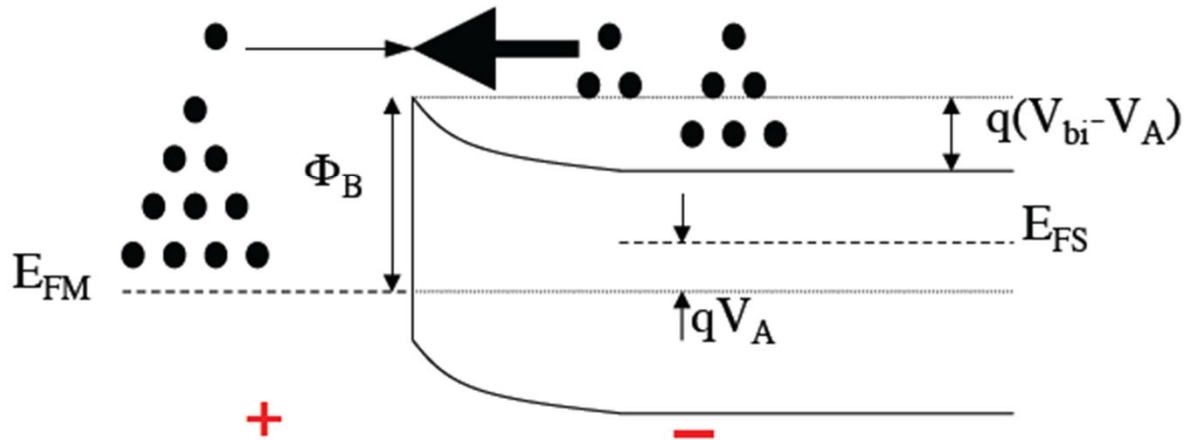
$$V_{bi} = \Phi_M - \Phi_S$$



Suppose an external potential is applied to the semiconductor. Since the impedance of the depletion region is much greater than the metal or the bulk semiconductor ($n \approx p \approx 0$ in depletion region), most of the voltage appears across the depletion region.

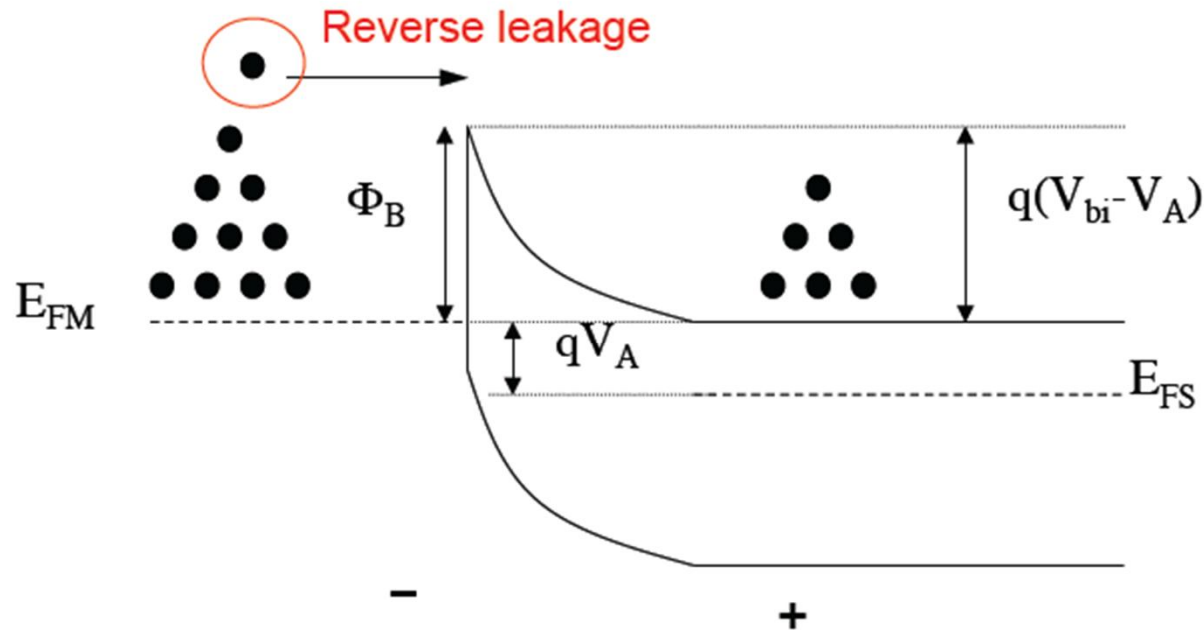
Schottky barrier diode - forward bias

Forward Bias



Net electron flow is from the semiconductor to metal since Φ_B , the barrier to electrons flowing into the semiconductor remains the same, but the effective barrier for electrons in the semiconductor is decreased to $V_{bi} - V_A < V_{bi}$.

Schottky barrier diode – reverse bias



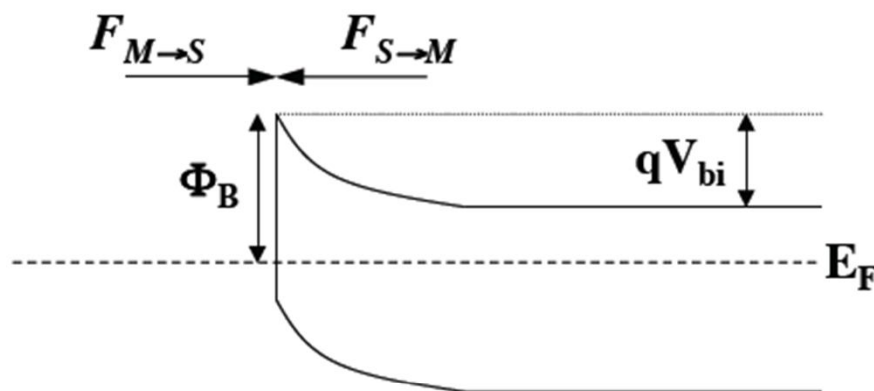
There is now net electron flow from metal to semiconductor because Φ_B remains the same, but the effective semiconductor barrier increases, $V_{bi} - V_A > V_{bi}$. Since $\Phi_B \gg kT$, there is very little current flow.

Schottky barrier diode

Note that E_F is not drawn in the depletion region under forward or reverse bias since this is a non-equilibrium condition.

A rectifying diode of the type described is often called a Schottky barrier diode after W. Schottky who first developed its theory in 1938. The theory was based upon work function differences, which were in error due to surface state pinning, but it still correctly explains barriers dominated by surface states by simply using the correct Φ_B , which is not equal to Φ_{MS} .

Current–Voltage Relationship



Under equilibrium conditions, the flow of electrons in each direction is the same. Under applied bias, $F_{S \rightarrow M}$ is changed by changing the barrier height $V_{bi} - V_A$. We cannot change $F_{M \rightarrow S}$ since no electric field can be supported in a perfect conductor.

Current is limited by either carriers surmounting the barrier (Thermionic Emission) or by carrier diffusion toward or away from the barrier (Diffusion Limited Transport) which resembles the situation in pn junctions. Thermionic Emission generally dominates unless the carrier mobility is very low.

Thermionic emission current

The current density $J_{S \rightarrow M}$ is a function of the number of electrons which have an x-directed velocity greater than the built-in potential.

$$J_{S \rightarrow M} = q \int_{qV_{bi}}^{\infty} v_x dn \qquad dn = g_c(E) f_F(E) dE$$

If all of the energy above the conduction band is kinetic energy,

$$\frac{1}{2} m^* v^2 = E - E_c = q(V_{bi} - V_A) \quad dn = \frac{4\pi (2m_n^*)^{3/2}}{h^3} \sqrt{E - E_c} \exp\left[\frac{-(E - E_F)}{kT}\right] dE$$

Using $J = J_{S \rightarrow M} - J_{M \rightarrow S}$ and that at equilibrium $J_{S \rightarrow M} = J_{M \rightarrow S}$

$$J = A^* T^2 \exp\left(\frac{-\Phi_B}{kT}\right) \exp\left(\frac{qV_A}{kT}\right) \qquad (13)$$

Where A^* is known as the Richardson constant $A^* \equiv \frac{4\pi q m_n^* k^2}{h^3}$

Thermionic emission current

$$J_{S \rightarrow M} = q \int_{qV_{bi}}^{\infty} v_x dn$$

$$dn = g(E)f(E)dE$$

$$dn = 4\pi \frac{(2m_n^*)^{\frac{3}{2}}}{h^3} (E - E_c)^{\frac{1}{2}} e^{-\frac{E - E_F}{k_0 T}} dE$$

$$= 4\pi \frac{(2m_n^*)^{\frac{3}{2}}}{h^3} \underbrace{e^{-\frac{E_c - E_F}{k_0 T}}}_{\text{constant}} \underbrace{(E - E_c)^{\frac{1}{2}} e^{-\frac{E - E_c}{k_0 T}}}_{\text{variable}} dE$$

$$\text{又: } E - E_c = \frac{1}{2} m_n^* v^2$$

$$\Rightarrow dE = m_n^* v dv$$

Thermionic emission current

$$\begin{aligned} n_0 &= N_c e^{-\frac{E_c - E_F}{k_0 T}} \\ &= \left[2 \frac{(2\pi m_n^* k_0 T)^{\frac{3}{2}}}{h^3} \right] \cdot e^{-\frac{E_c - E_F}{k_0 T}} \end{aligned}$$

$$\Rightarrow dn = n_0 \left(\frac{m_n^*}{2\pi k_0 T} \right)^{\frac{3}{2}} \cdot e^{-\frac{m_n^* v^2}{2k_0 T}} dv$$

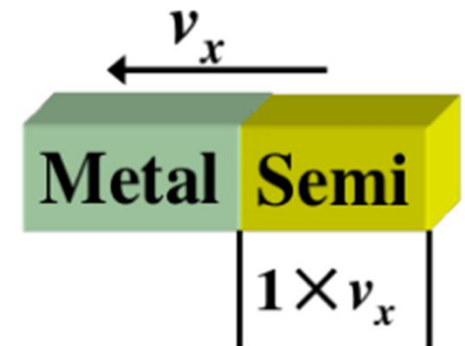
$$= n_0 \left(\frac{m_n^*}{2\pi k_0 T} \right)^{\frac{3}{2}} \cdot e^{-\frac{m_n^* (v_x^2 + v_y^2 + v_z^2)}{2k_0 T}} dv_x dv_y dv_z$$

Thermionic emission current

v_x 正方向为垂直于半导体指向金属界面的方向。

单位时间，通过单位截面积，在 $1 \times 1 \times v_x$ 体积内的电子可到达界面

$$dN = dn \cdot (v_x \times 1 \text{秒} \times 1 \text{单位截面})$$



$$= n_0 \left(\frac{m_n^*}{2\pi k_0 T} \right)^{\frac{3}{2}} \cdot e^{-\frac{m_n^* (v_x^2 + v_y^2 + v_z^2)}{2k_0 T}} \cdot v_x \cdot dv_x dv_y dv_z$$

要越过势垒，到达界面的电子的动能

$$\frac{1}{2} m_n^* v_x^2 \geq q(V_{bi} - V_A)$$

Thermionic emission current

即电子的最小速度 $v_x(\min) = \sqrt{\frac{2q(V_{bi} - V_A)}{m_n^*}}$

电子流密度:

$$J_{s \rightarrow m} = \int q v_x dn$$

$$= q n_0 \left(\frac{m_n^*}{2\pi k_0 T} \right) \cdot \int_{-\infty}^{+\infty} dv_z \cdot \int_{-\infty}^{+\infty} dv_y \cdot \int_{v_{x0}}^{+\infty} v_x e^{-\frac{m_n^* (v_x^2 + v_y^2 + v_z^2)}{2k_0 T}} dv_x$$

$$= A^* T^2 e^{\left(-\frac{q\phi_{ns}}{k_0 T} \right)} e^{\frac{qV}{k_0 T}}$$

其中, $A^* = \frac{4\pi q m_n^* k_0^2}{h^3}$ 有效理查逊常数

Thermionic emission current

Using $J = J_{S \rightarrow M} - J_{M \rightarrow S}$ and that at equilibrium $J_{S \rightarrow M} = J_{M \rightarrow S}$

$$J_{m \rightarrow s} = -A^* T^2 e^{\left(-\frac{q\phi_{ns}}{k_0 T} \right)}$$

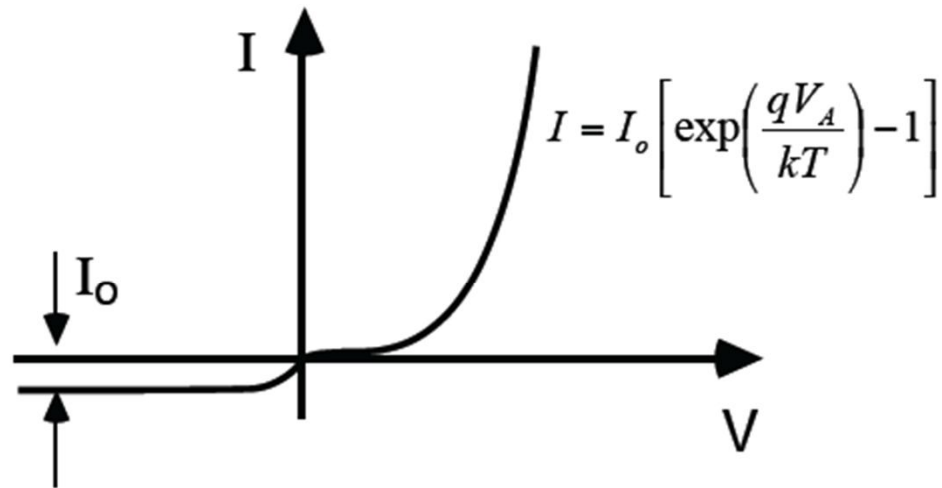
$$J = J_{s \longrightarrow m} + J_{m \longrightarrow s}$$

$$= J_{ST} \left(e^{\frac{qV}{k_0 T}} - 1 \right)$$

$$\text{其中 } J_{ST} = A^* T^2 e^{\left(-\frac{q\phi_{ns}}{k_0 T} \right)}$$

Φ_{ns} 是金属一边的电子势垒

Current–Voltage Relationship



With forward bias, I increases exponentially with V_A

With reverse bias, $I \approx I_o$ (constant)

While the I-V characteristic “looks like” the ideal pn diode, the current mechanisms are entirely different

Nonideal Effects on the Barrier Height

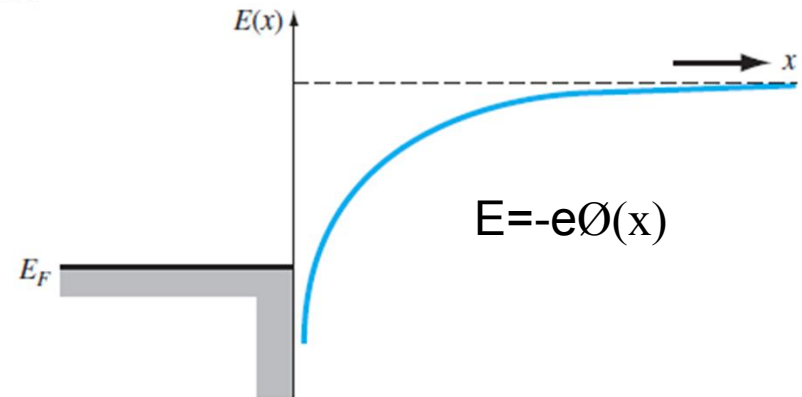
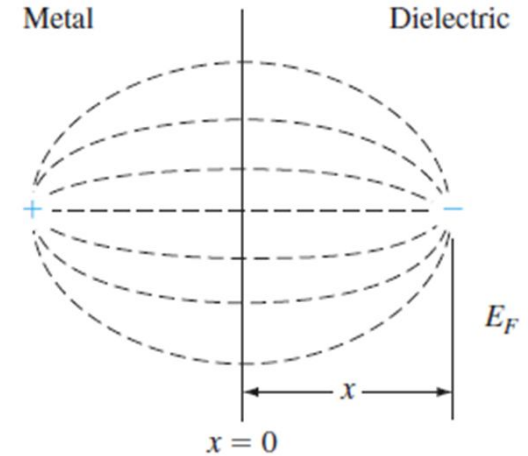
- Schottky effect, or **image-force-induced lowering of the potential barrier**.
- An electron in a dielectric material at a distance x from the metal will create an electric field.
- The electric field can be determined by adding an image charge, e , inside the metal located at the same distance, x , from the interface.

$$F = \frac{-e^2}{4\pi\epsilon_s(2x)^2} = -eE$$

- The potential can then be found as:

$$-\phi(x) = + \int_x^\infty E dx'$$

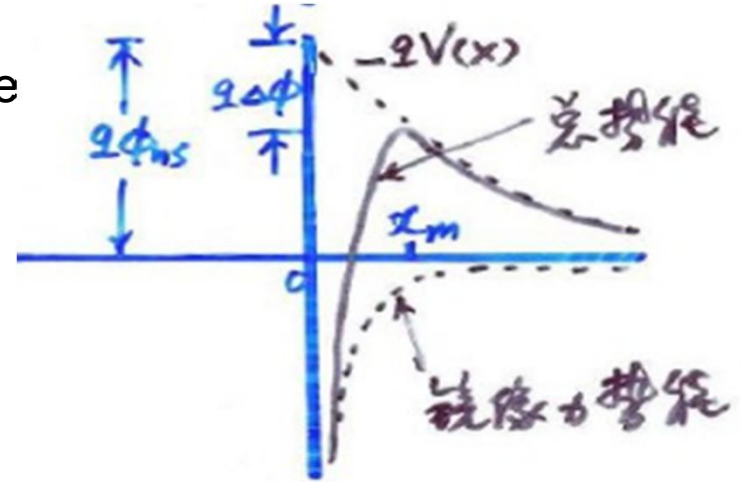
$$= + \int_x^\infty \frac{e}{4\pi\epsilon_s \cdot 4(x')^2} dx' = \frac{-e}{16\pi\epsilon_s x}$$



Nonideal Effects on the Barrier Height

With an electric field present in the dielectric, the potential is modified and can be written as

$$-\phi(x) = \frac{-e}{16\pi\epsilon_s x} - Ex$$



We can find the Schottky barrier lowering, $\Delta\phi$, and the position of the maximum barrier, x_m , from the condition that

$$\frac{d[e\phi(x)]}{dx} = 0$$

We find that

$$x_m = \sqrt{\frac{e}{16\pi\epsilon_s E}}$$

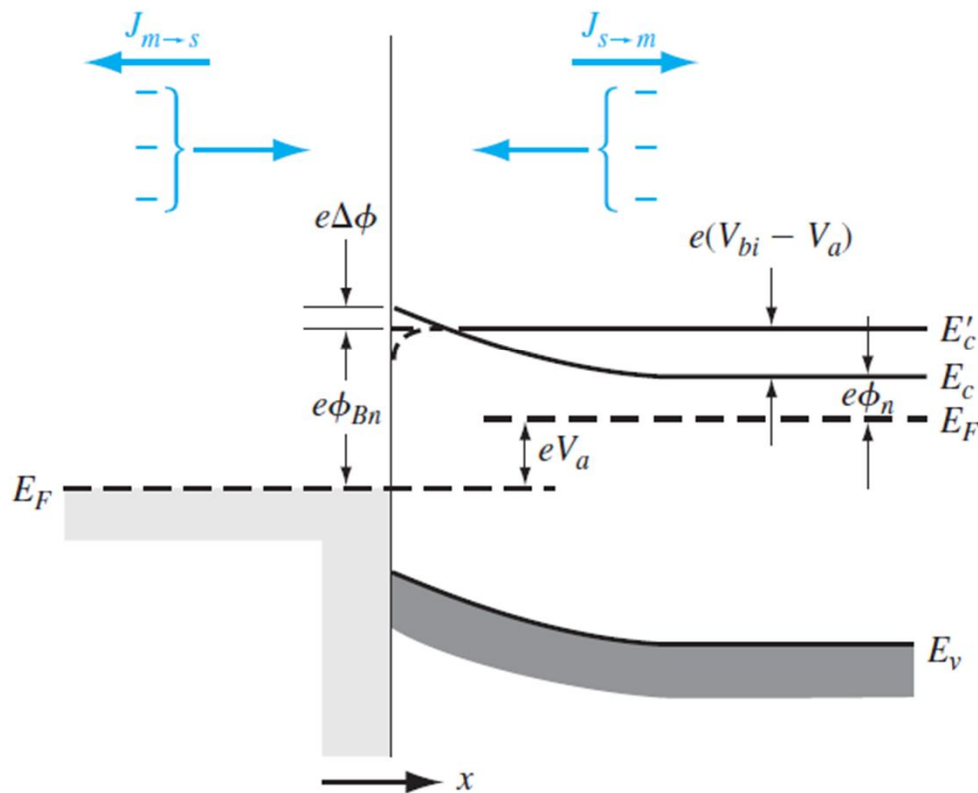
and

$$\Delta\phi = \sqrt{\frac{eE}{4\pi\epsilon_s}}$$

The Schottky barrier height is a function of the electric field through the barrier lowering effect.

Although the Schottky barrier lowering may seem like a small value, it can have a significant effect on the current in a Schottky barrier diode.

Thermionic emission current (image-force-induced lowering effect)

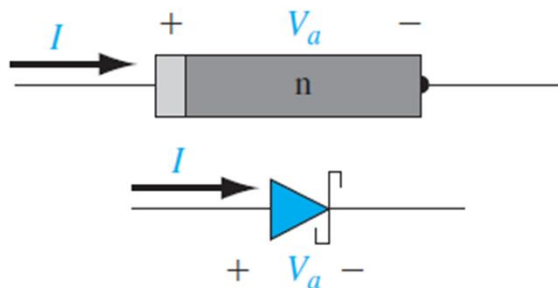


$$J = \left[A^* T^2 \exp \left(\frac{-e\phi_{Bn}}{kT} \right) \right] \left[\exp \left(\frac{eV_a}{kT} \right) - 1 \right]$$

$$A^* \equiv \frac{4\pi e m_n^* k^2}{h^3}$$

$$J = J_{sT} \left[\exp \left(\frac{eV_a}{kT} \right) - 1 \right]$$

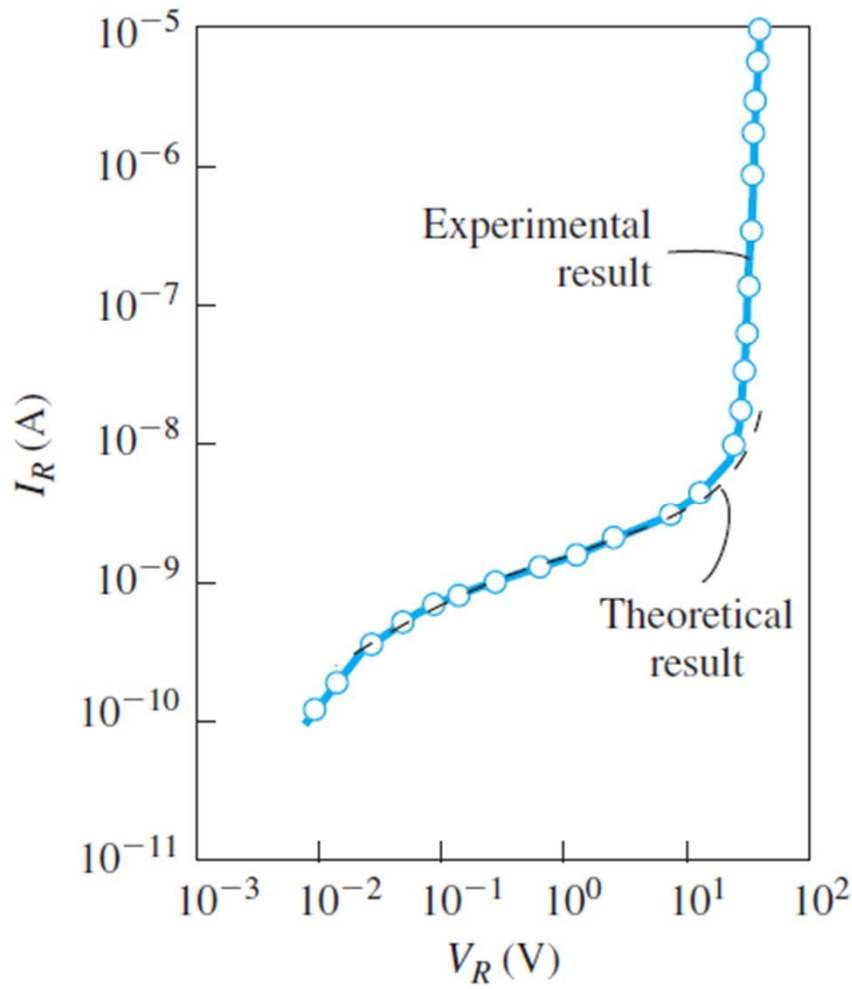
$$J_{sT} = A^* T^2 \exp \left(\frac{-e\phi_{Bn}}{kT} \right)$$



Considering image-force-induced lowering of the potential barrier:

$$J_{sT} = A^* T^2 \exp \left(\frac{-e\phi_{B0}}{kT} \right) \exp \left(\frac{e\Delta\phi}{kT} \right)$$

Thermionic emission current (image-force-induced lowering effect)



$$\Delta\phi = \sqrt{\frac{eE}{4\pi\epsilon_s}}$$

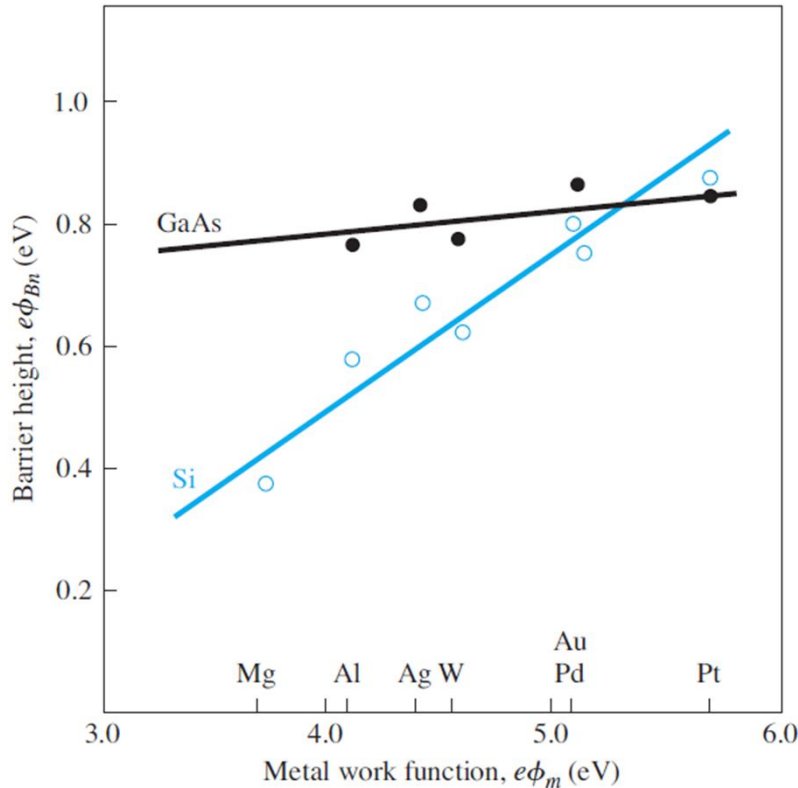
$$J_{sT} = A^*T^2 \exp\left(\frac{-e\phi_{B0}}{kT}\right) \exp\left(\frac{e\Delta\phi}{kT}\right)$$

The change in barrier height, will increase with an increase in the electric field, or with an increase in the applied reverse-biased voltage.

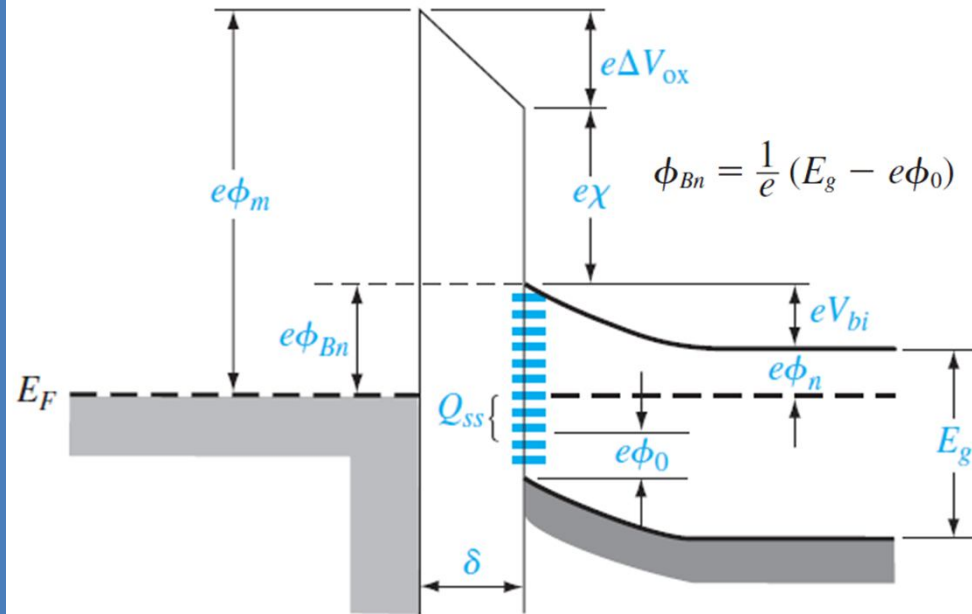
The reverse biased current increases with reverse-biased voltage because of the barrier lowering effect. This figure also shows the Schottky barrier diode going into breakdown.

Nonideal Effects on the Barrier Height

Interface States

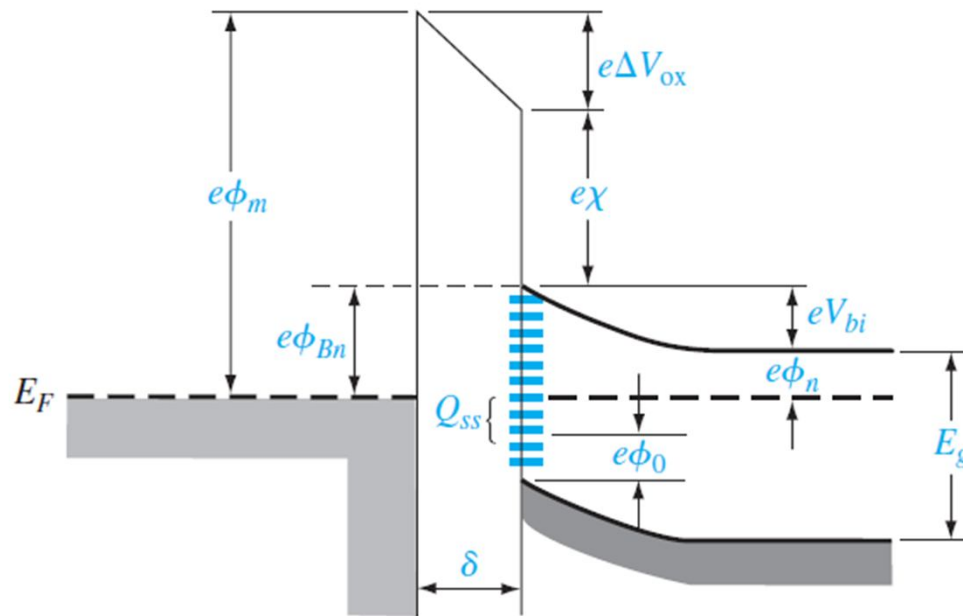


The barrier height of the metal-semiconductor junction is determined by both the metal work function and the semiconductor surface or interface states.



- a narrow interfacial layer of insulator exists between the metal and semiconductor, which can support a potential difference
- Since the surface state density is not predictable with any degree of certainty, the barrier height must be an experimentally determined parameter.

Nonideal Effects on the Barrier Height



$$\phi_{Bn} = \frac{1}{e} (E_g - e\phi_0)$$

all states below the surface potential ϕ_0 are donor states, above ϕ_0 are acceptor states. It seems that the Fermi level is fixed at ϕ_0 .

The barrier height is now fixed by the bandgap energy and the potential ϕ_0 . The barrier height is totally independent of the metal work function and the semiconductor electron affinity. The Fermi level becomes “pinned” at the surface, at the surface potential ϕ_0 .

Diffusion Limited Current

For $W > L$, there are multiple scatterings during electron movement, and thus the electron velocity cannot increase unlimitedly. We should use the diffusion limit transport to derive the J-V.

$$F_{S \rightarrow M} = D_N \frac{dn}{dx} + n\mu E$$

By applying an external potential, E can be changed. Under forward bias E is reduced, therefore the drift term is decreased, allowing diffusion to produce a net flux or current.

The number of electrons with sufficient energy to surmount the respective barriers qV_{bi} and Φ_B are given by Fermi-Dirac statistics (or Maxwell-Boltzmann approximation).

$$n_{S-M} = n \exp\left(-\frac{qV_{bi}}{kT}\right)$$

Note that the number of electrons in the metal $\gg N_D$.

Hence, the barrier, Φ_B , must be higher than qV_{bi} to get the same flux in both directions at equilibrium.

Diffusion Limited Current

In equilibrium $I_{S \rightarrow M} = I_{M \rightarrow S} = -Kne^{-\frac{qV_{bi}}{kT}}$ Where K is a proportionality factor

Under bias, $V_{bi} \rightarrow (V_{bi} - V_A)$ where $V_A =$ applied bias

$$n_{S-M} = ne^{-\frac{q(V_{bi} - V_A)}{kT}}$$

Since the current flowing $M \rightarrow S$ is unchanged by V_A ,

$$\begin{aligned} I &= I_{M \rightarrow S} - I_{S \rightarrow M} \\ &= -KN_D e^{-\frac{qV_{bi}}{kT}} + KN_D e^{-\frac{q(V_{bi} - V_A)}{kT}} \end{aligned}$$

$$I = I_o \left[\exp\left(\frac{qV_A}{kT}\right) - 1 \right]$$

where $I_o = KN_D e^{-\frac{qV_{bi}}{kT}}$ is reverse saturation current

Comparison of the Schottky Barrier Diode and the pn Junction Diode

there are two important differences between a Schottky diode and a pn junction diode:

- The first is in the magnitudes of the reverse - saturation current densities
- The second is in the switching characteristics.

The reverse-saturation current density of the Schottky barrier diode was given by Equation (9.26) and is

$$J_{sT} = A^* T^2 \exp\left(\frac{-e\phi_{Bn}}{kT}\right)$$

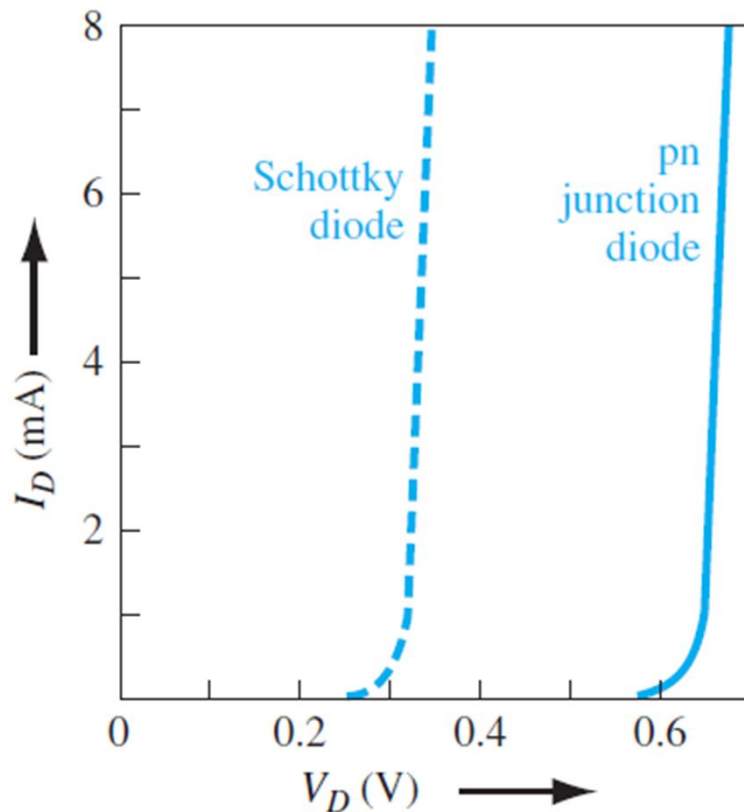
The ideal reverse-saturation current density of the pn junction diode can be written as

$$J_s = \frac{eD_n n_{po}}{L_n} + \frac{eD_p p_{no}}{L_p} \quad (9.28)$$

The form of the two equations is vastly different, and the current mechanism in the two devices is different. The current in a pn junction is determined by the diffusion of minority carriers while the current in a Schottky barrier diode is determined by thermionic emission of majority carriers over a potential barrier.

The ideal reverse-saturation current density of the Schottky barrier junction is orders of magnitude larger than that of the ideal pn junction diode.

Comparison of the Schottky Barrier Diode and the pn Junction Diode

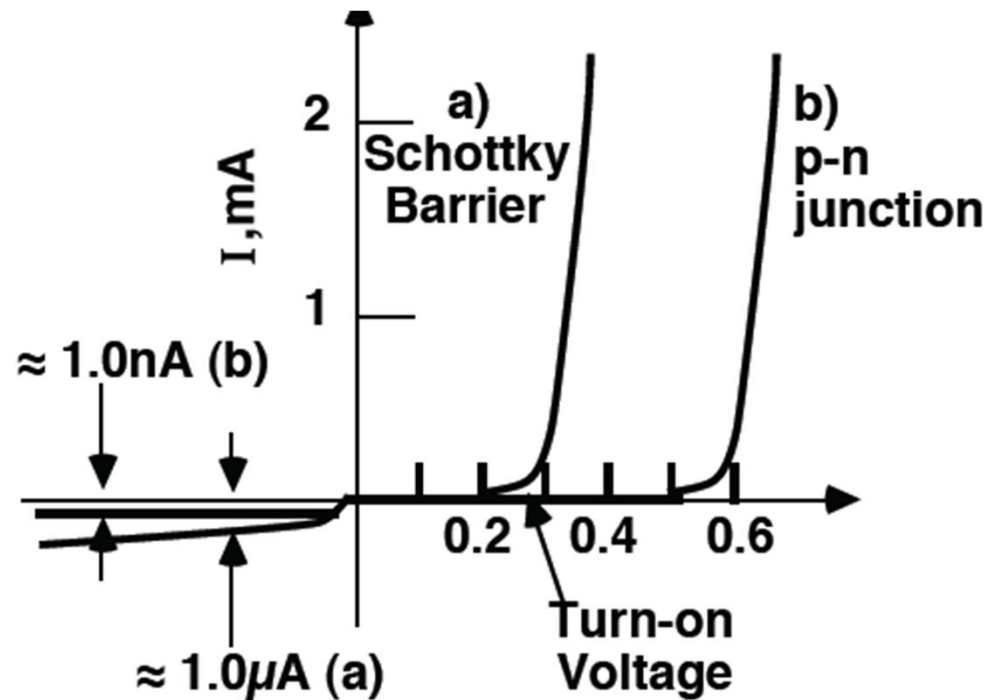


Since $J_s T > J_s$, the forward - bias characteristics of the two types of diodes will also be different.

The second major difference is in the frequency response

- there is no diffusion capacitance associated with a forward - biased Schottky diode. The elimination of the diffusion capacitance makes the Schottky diode a higher - frequency device than the pn junction diode.
- when switching a Schottky diode from forward to reverse bias, there is no minority carrier stored charge to remove, as is the case in the pn junction diode.

Comparison of the Schottky Barrier Diode and the pn Junction Diode



Compared with P-N junctions, Schottky diodes have

1. lower forward turn-on voltage
2. higher reverse saturation current
3. Faster switching due to no minority carrier charge storage
4. more sensitivity to surface preparation
5. lower reverse breakdown voltage.

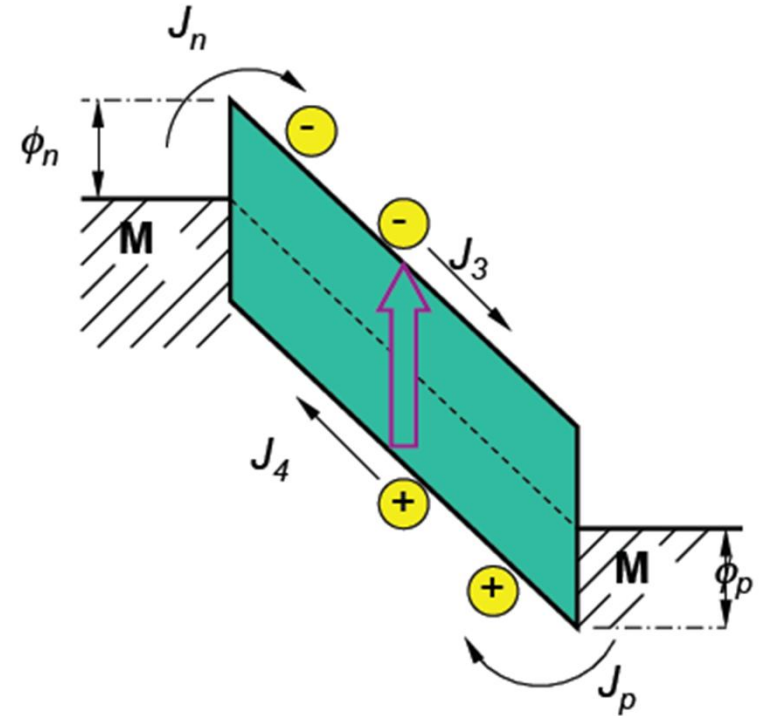
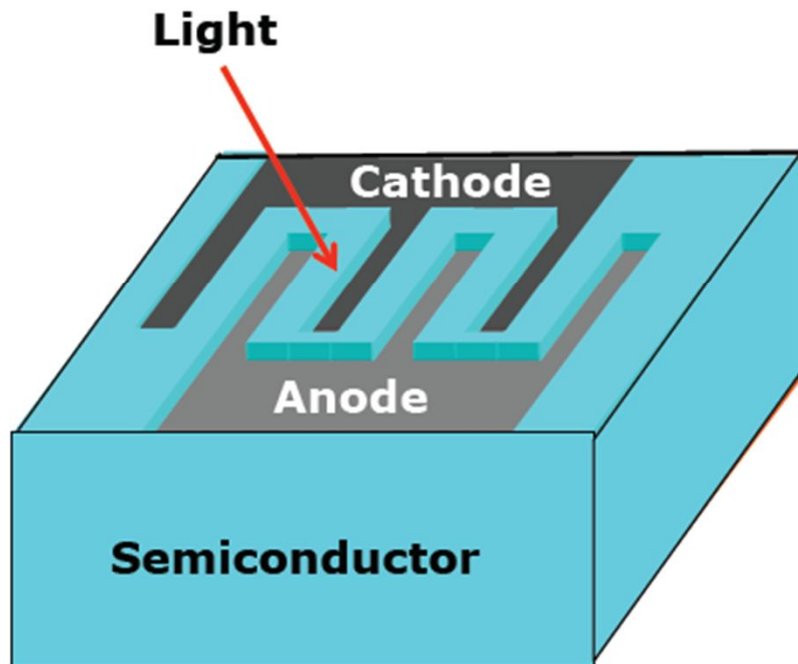
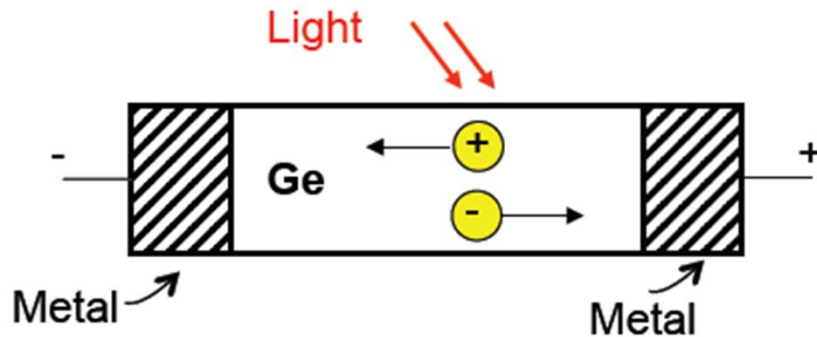
Schottky diode applications

Schottky diodes have found widespread application as fast switching diodes because they have no minority carrier charge storage.

- They have also been used in bipolar IC's where they are used as clamping diodes to prevent saturation and minority carrier storage in the collector region of NPN bipolar transistors.
- Schottky diodes are used as optical detectors and solar cells
- Most recently Schottky contacts have been used to form source/drain regions of MOS transistors.

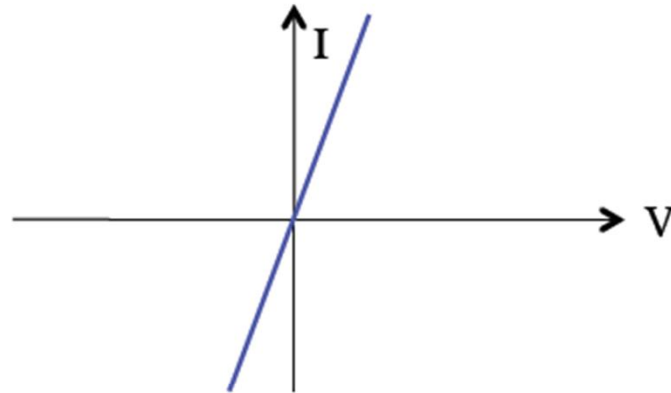
Schottky contact MSM photodetector

Schottky contacts used to collect optically generated electrons and holes



- Very simple technology
- Low thermal budget processing
- Low cost
- Applications
 - Optical interconnects
 - Night vision

Ohmic contacts

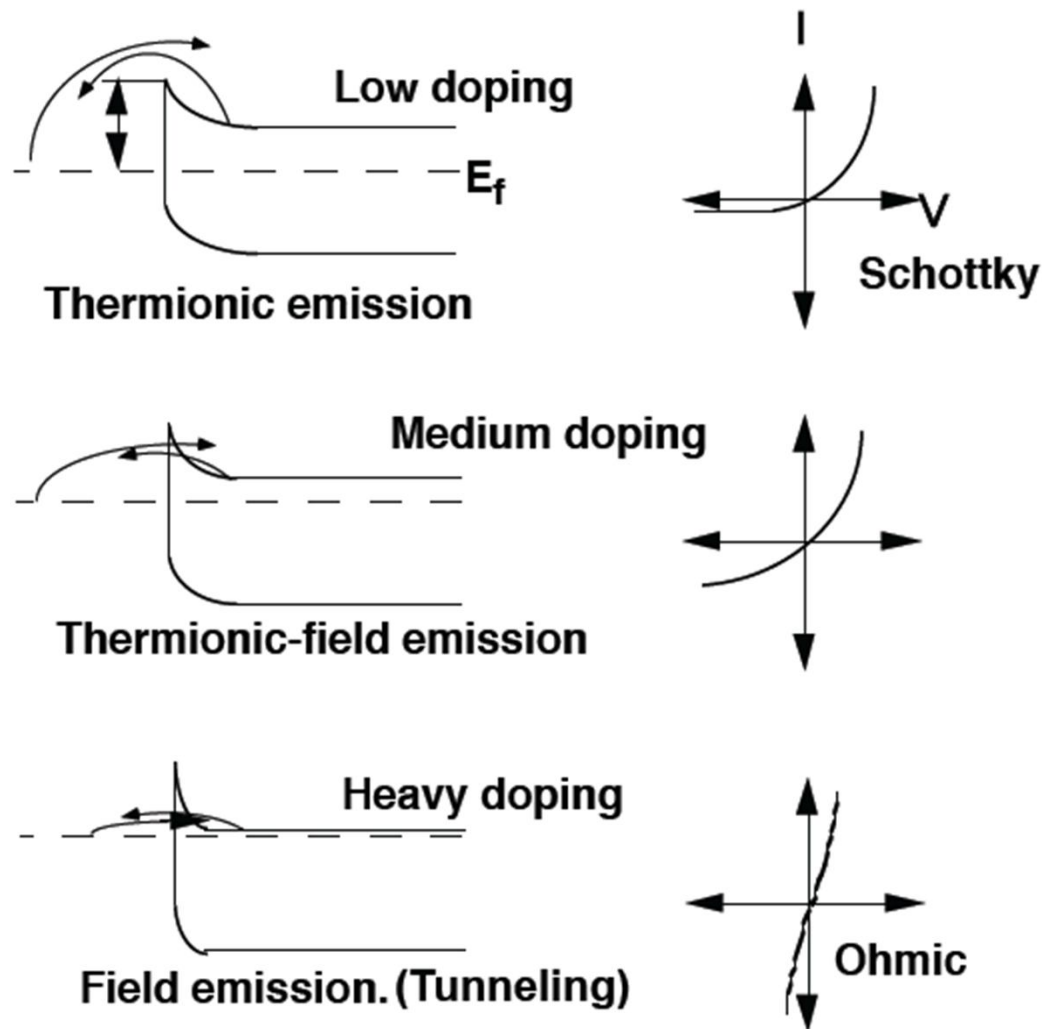


An Ohmic contact is defined as one in which there is an unimpeded transfer of majority carriers from one material to another, i.e., the contacts do not limit the current.

There are two basic ways to achieve such a contact:

- A) Dope the semiconductor heavily enough that tunneling is possible.
- B) Choose a system where Φ_M and Φ_S are such that there is no barrier to current flow

Conduction mechanisms in M/S contacts



Conduction strongly depends on barrier height (ϕ_B) and doping density

Conduction mechanisms in M/S contacts

Suppose N_D or N_A in the semiconductor is very large.

Then $W = \sqrt{\frac{2K_s \epsilon_o V_{bi}}{qN_{D,A}}}$ becomes very small

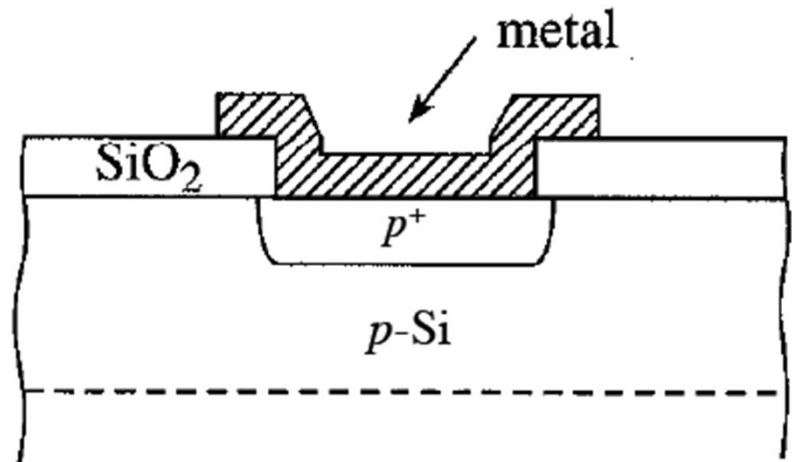
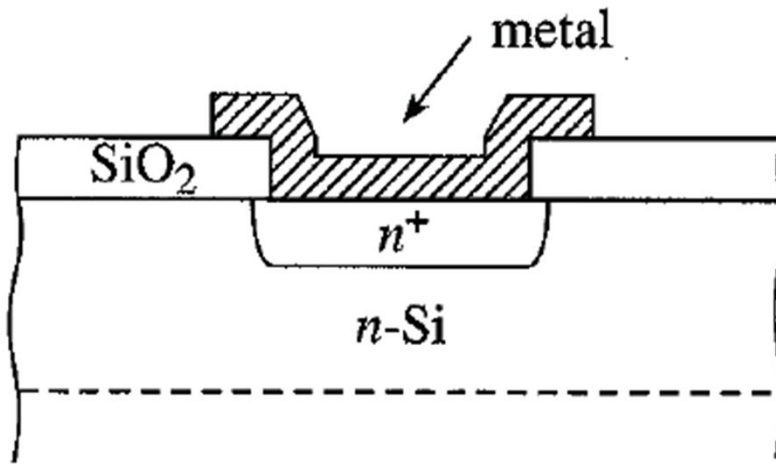
When $W \approx 25\text{--}50 \text{ \AA}$, electrons can “tunnel” through the barrier. This process occurs in both directions $M \rightarrow S$ and $S \rightarrow M$ so the contact shows very little resistance in either direction and becomes Ohmic.

To calculate an approximate value for the required doping,

$$N_{D_{\min}} \approx \frac{2K_s \epsilon_o V_{bi}}{qW^2} \approx 6.2 \times 10^{19} \text{ cm}^{-3} \text{ for } W = 25 \text{ \AA}$$

This is a relatively easy value to achieve in practice and is normally how Ohmic contacts are made in integrated circuits.

Tunneling ohmic contact

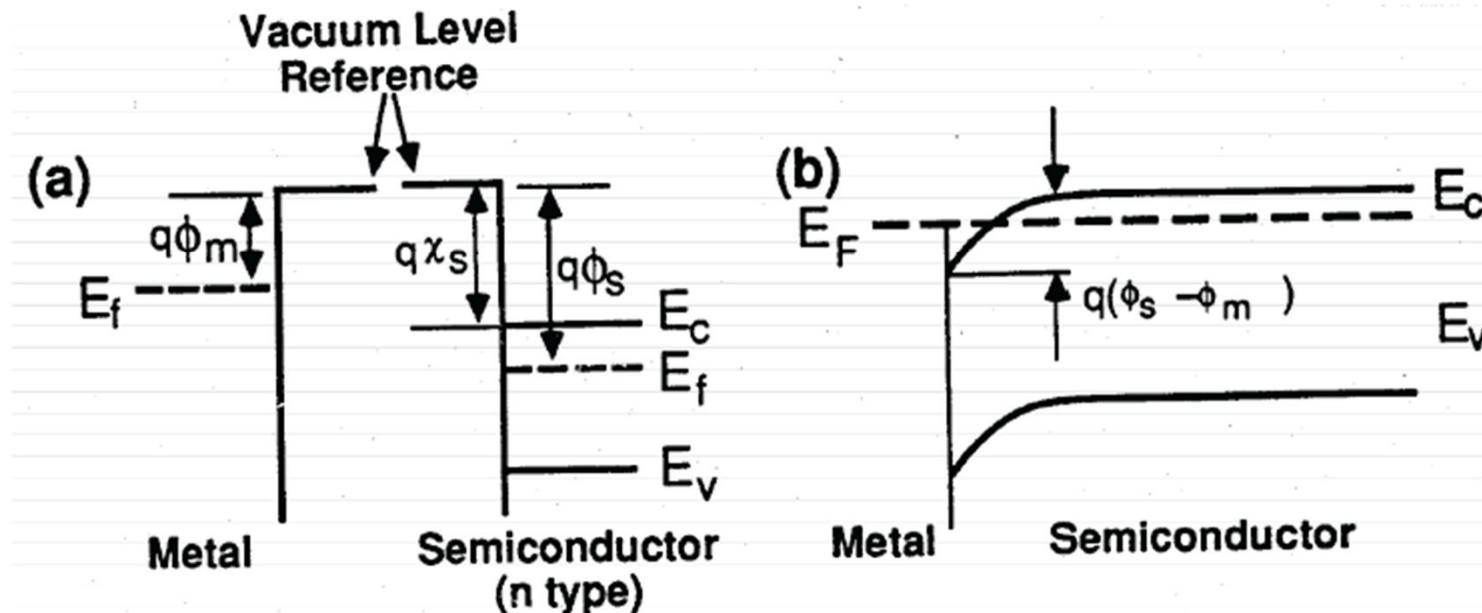


Contacts are typically heavily doped (n^+ or p^+) to insure Ohmic contact formation

Contacts without appreciable barrier

Alternatively, one can try to find a metal such that $\Phi_M < \Phi_S$ for N-type, or $\Phi_M > \Phi_S$ for P-type.

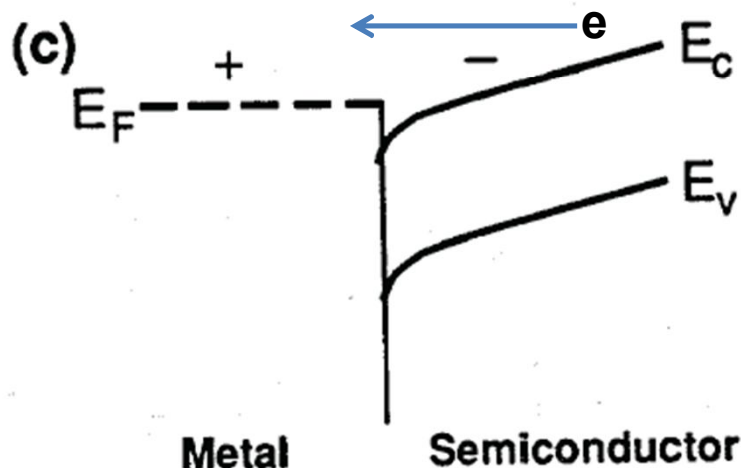
Consider an N-type semiconductor:



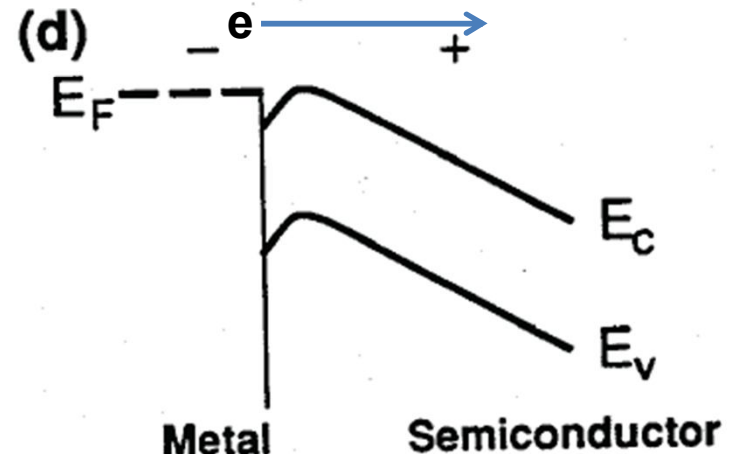
Contacts without appreciable barrier

If $\Phi_M < \Phi_S$, then equilibrium is established by electrons flowing from M $\rightarrow$ S. **An Ohmic contact results because the semiconductor surface is not depleted when the metal comes into contact with the semiconductor.** Rather, there is an accumulation of majority carriers at the surface. In general, if majority carriers are enhanced near the surface, the contact will be Ohmic.

Metal biased positive

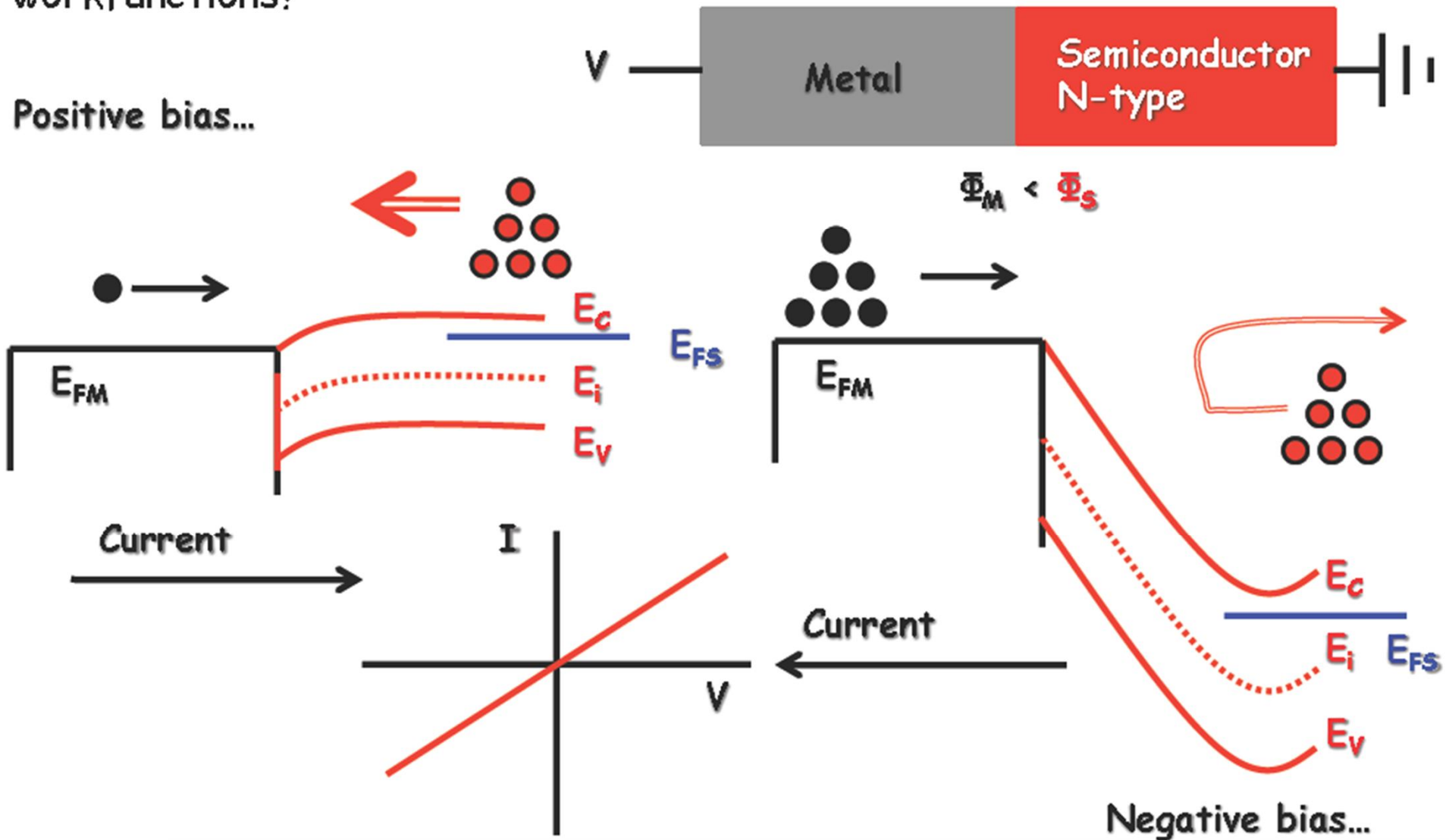


Metal biased negative



Ohmic contacts (n-type)

What happens when we reverse the relationship between the workfunctions?



Ohmic contacts (p-type)

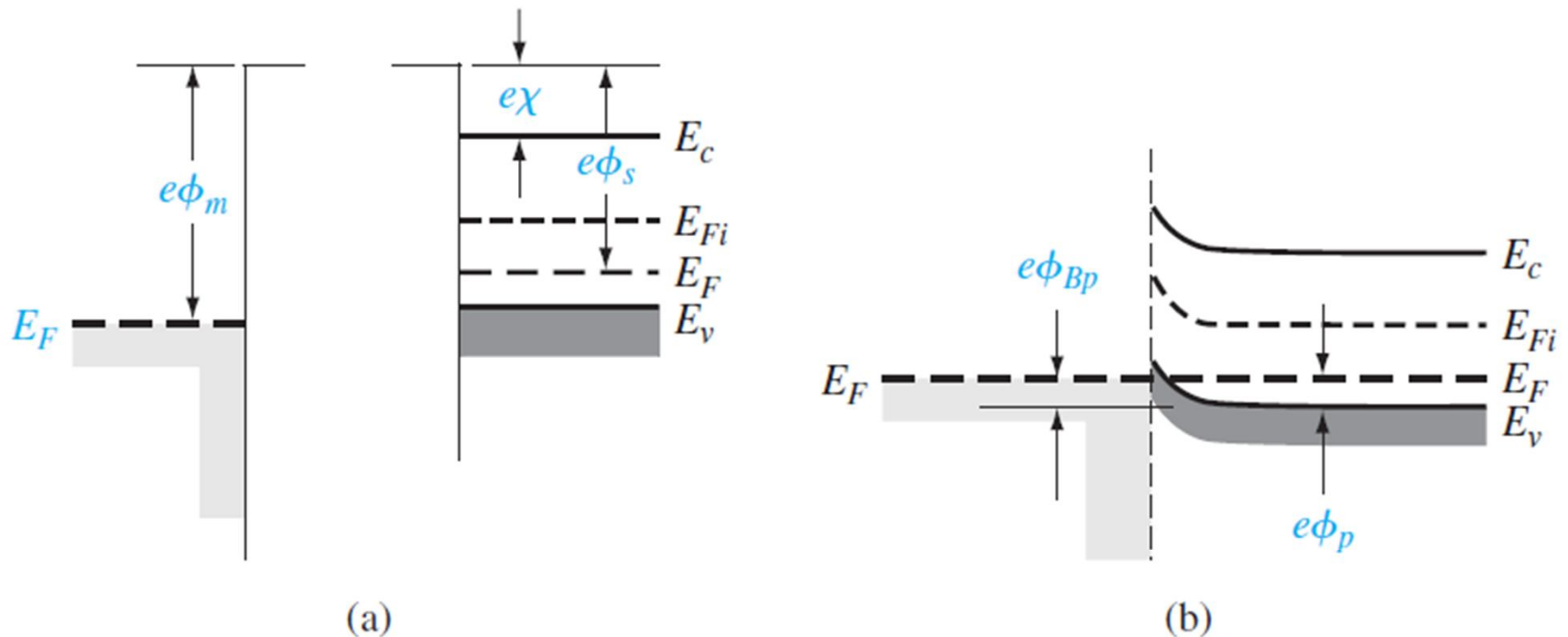
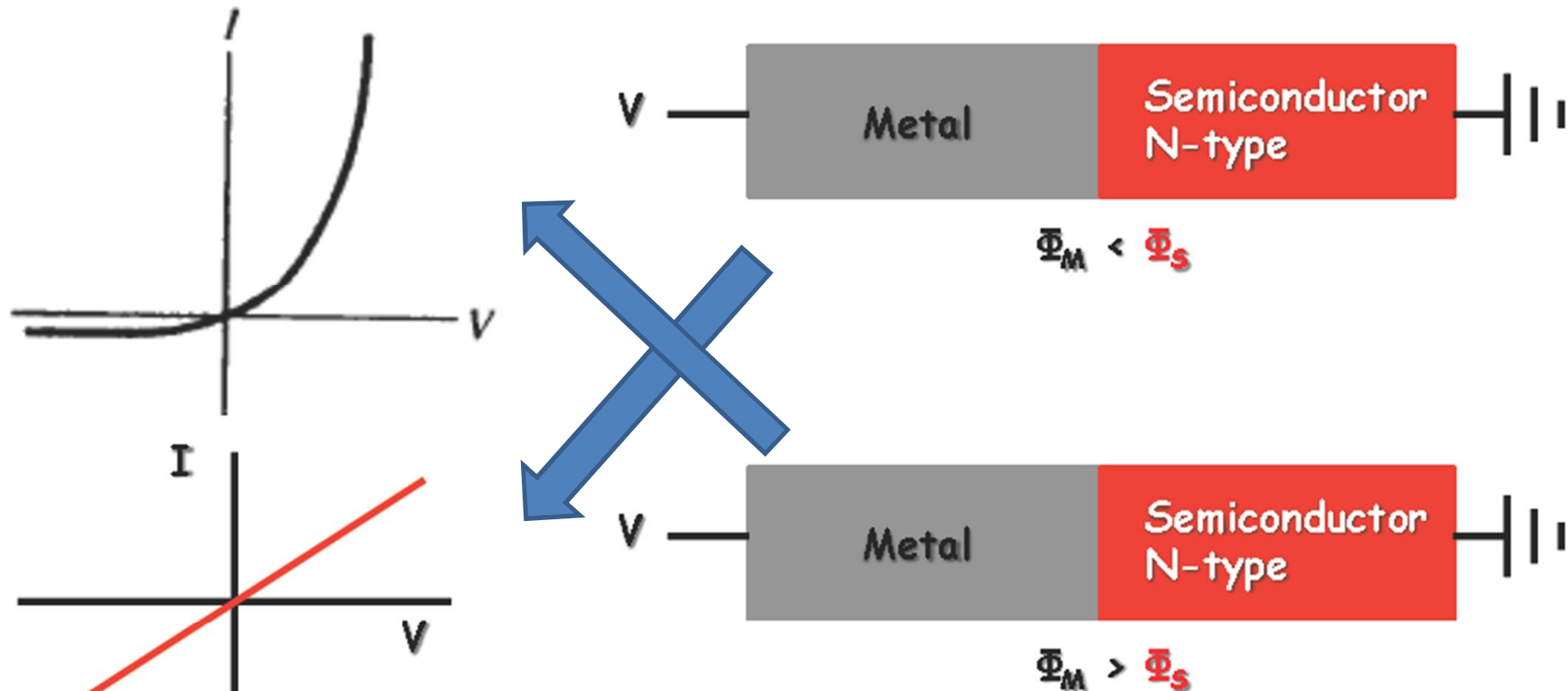


Figure 9.13 | Ideal energy-band diagram (a) before contact and (b) after contact for a metal–p-type semiconductor junction for $\phi_m > \phi_s$.

Ohmic contacts

Let's summarize what we have so far...



	N-type semiconductor	P-type semiconductor
$\phi_M > \phi_S$	Rectifying	Ohmic
$\phi_M < \phi_S$	Ohmic	Rectifying

9.3 | HETEROJUNCTIONS

If both the p-type and the n-type regions are of the same semiconductor material, the junction is called a **homojunction**. If the junction layers are made of different semiconductor materials, it is a **heterojunction**.

As a matter of convention, if the n-type doped semiconductor material has larger energy gap than the p-type doped material, it is denoted a p-N heterojunction.

The use of capital and lowercase letters connotes the relative size the energy gap. Conversely, if the p-type doped material has a larger energy gap than the n-type material, the junction is referred to as a P-n heterojunction.

9.3 | HETEROJUNCTIONS

Type I (Straddling Alignment)

- AlGaAs/GaAs
- GaSb/AlSb
- GaAs/GaP

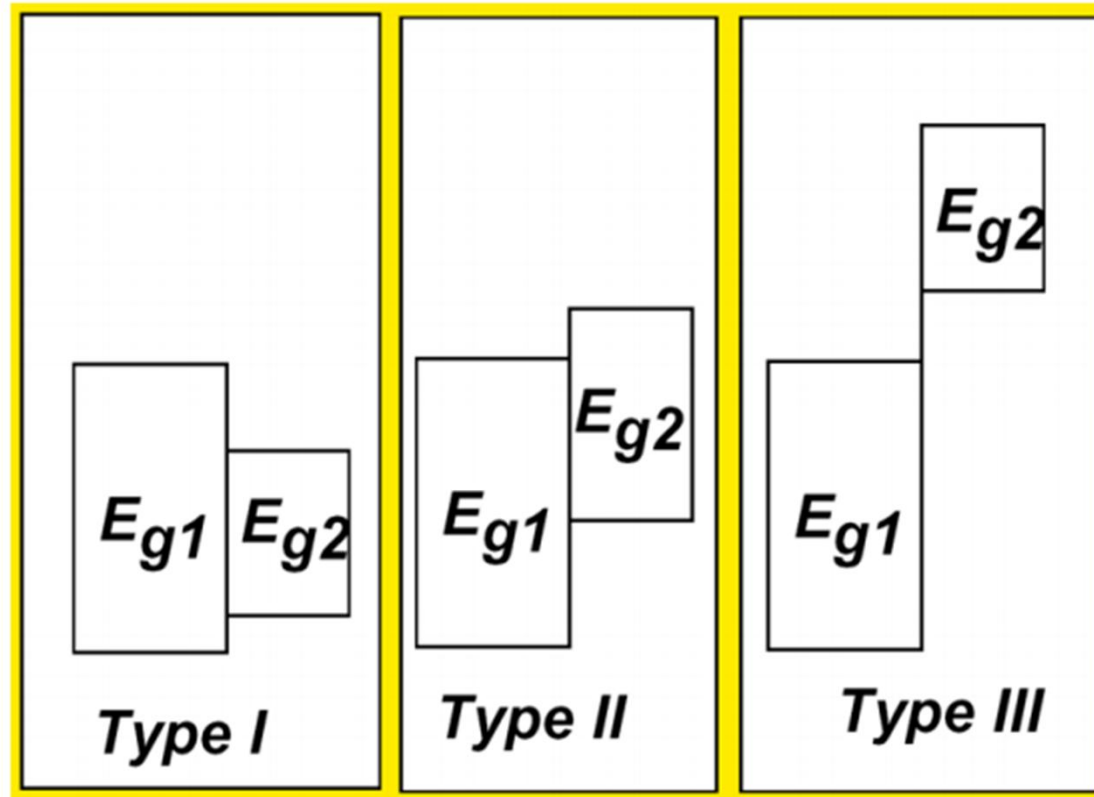
Type II (Staggered)

Either ΔE_c or ΔE_v is $> \Delta E_g$

- InP/ $\text{Al}_{0.48}\text{In}_{0.52}\text{P}$
- $\text{In}_x\text{Ga}_{1-x}\text{As}/\text{Ga}_x\text{Sb}_x\text{As}$
- $\text{Al}_x\text{In}_{1-x}\text{As}/\text{InP}$

Type III (broken gap)

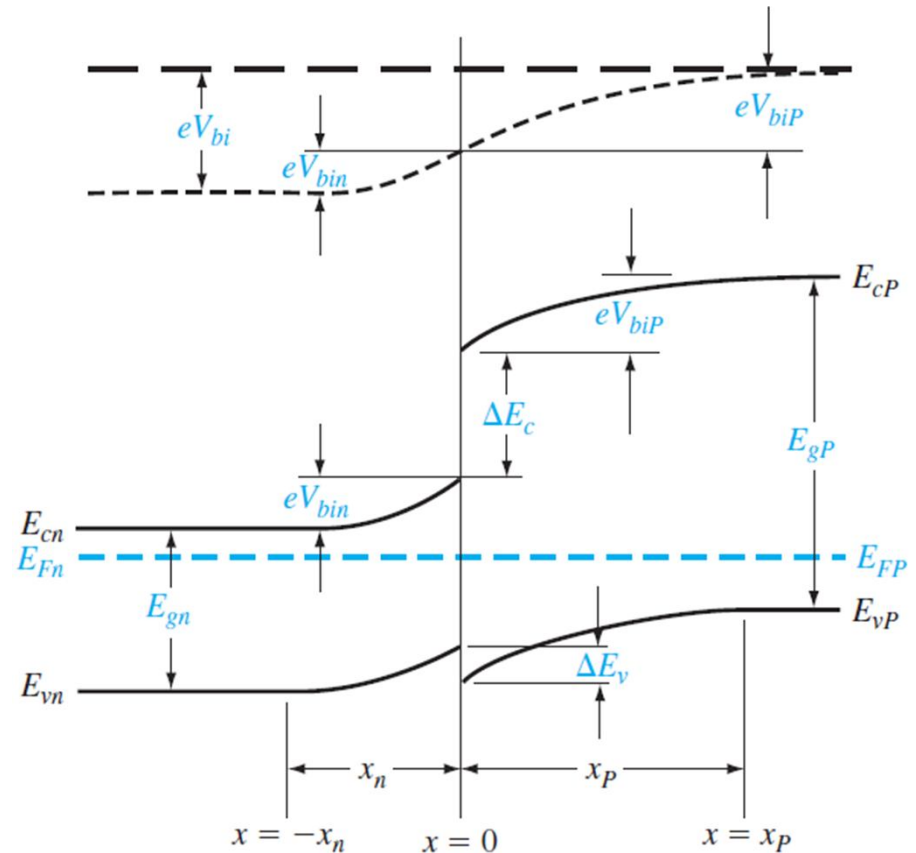
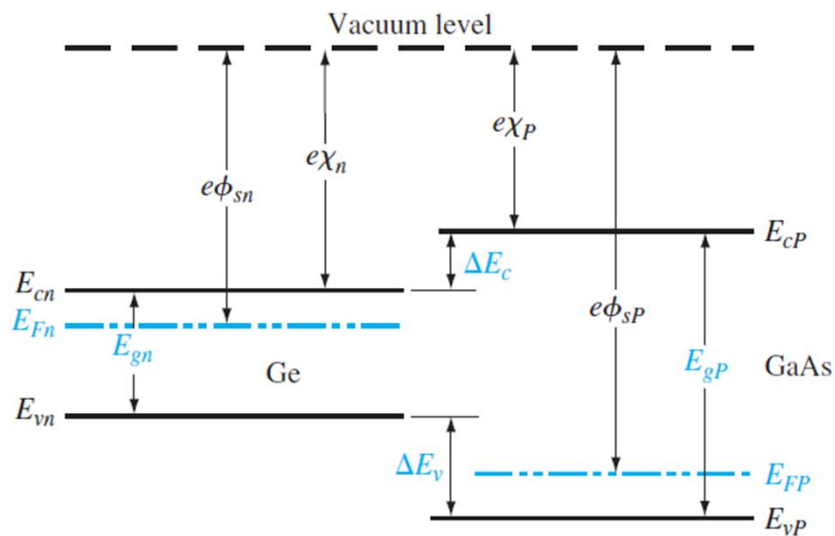
- InAs/GaSb



Heterojunction: Band lineup

$$\Delta E_c = e(\chi_n - \chi_p)$$

$$\Delta E_c + \Delta E_v = E_{gp} - E_{gn} = \Delta E_g$$



Heterojunction: Band lineup

- Determine band discontinuities ΔE_c and ΔE_v from difference in electron affinities:

$$\Delta E_c = E_{c2} - E_{c1} = \chi_1 - \chi_2$$

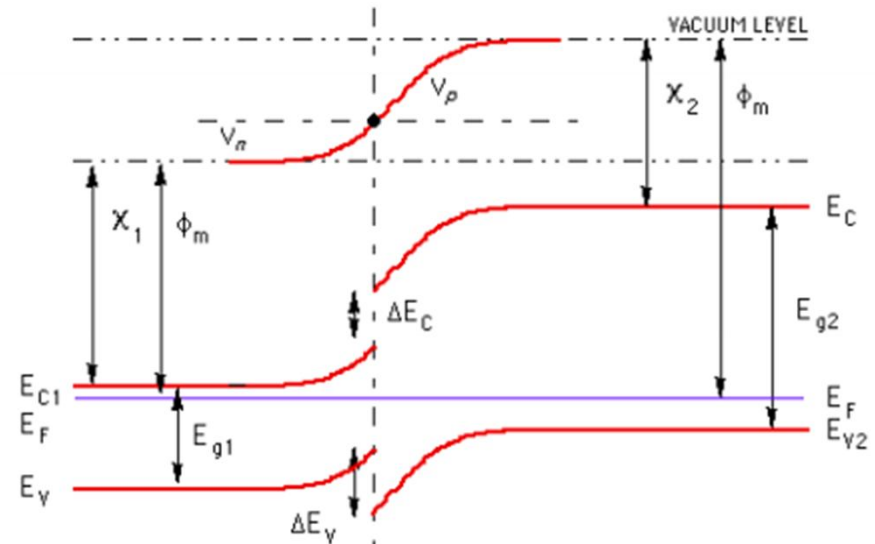
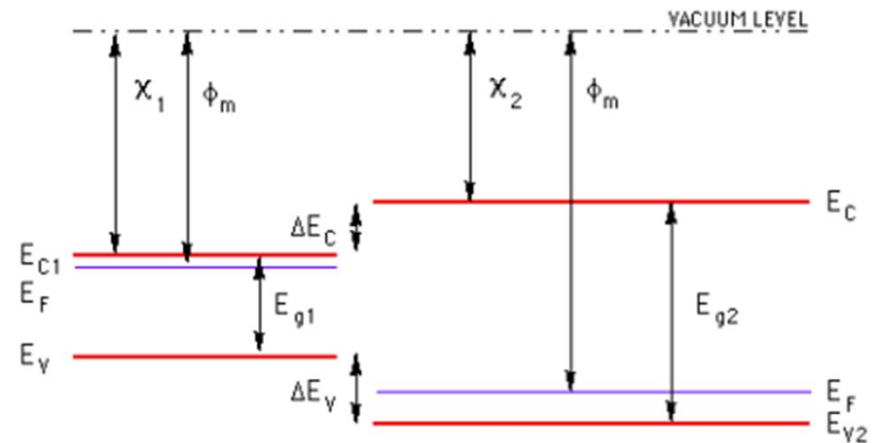
$$\Delta E_v = E_{v1} - E_{v2} = E_{g2} - E_{g1} - \Delta E_c = \Delta E_g - \Delta E_c$$

- Determine the built-in voltage from the difference in work functions.

$$q\phi_i = E_{Fn} - E_{Fp} = q\phi_{m1} - q\phi_{m2}$$

- Draw the bands at $\pm\infty$ for the appropriate doping
- Determine V_n and V_p and x_n and x_p from the solution of Poisson's Equation using Full Depletion Approximation with ϕ_i as a boundary condition.

Formation of P-n heterojunction



p-N heterojunction: electrostatics

Junction barrier:

$$q\phi_i = E_{Fn} - E_{Fp} = \Delta E_c + kT \ln \frac{n_{n0} N_{cp}}{n_{p0} N_{cn}}$$

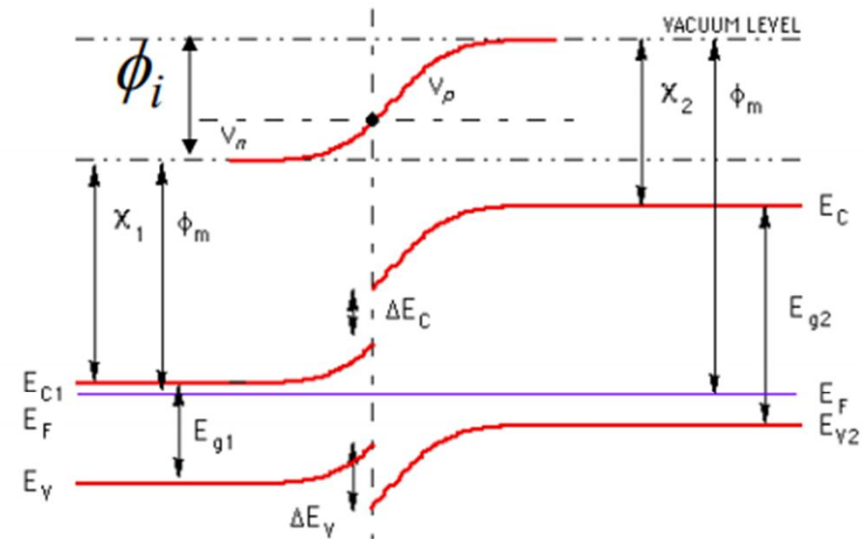
The same using valence band parameters

$$q\phi_i = -\Delta E_v + kT \ln \frac{p_{p0} N_{vn}}{p_{n0} N_{vp}}$$

- Or combining the two, independent of the free carrier concentrations:

$$q\phi_i = \frac{\Delta E_c - \Delta E_v}{2} + kT \ln \frac{N_d N_a}{n_{in} n_{ip}} + \frac{kT}{2} \ln \frac{N_{vn} N_{cp}}{N_{cn} N_{vp}}$$

Flat-band diagram of a p-n heterojunctions



p-N heterojunction: electrostatics

Depletion widths and band bending are calculated similar to a p-n homojunction by solving the Poisson equation:

$$x_n = \sqrt{\frac{2\epsilon_n}{q} \frac{(1-\xi)}{N_d} (\phi_i - V_a)}$$

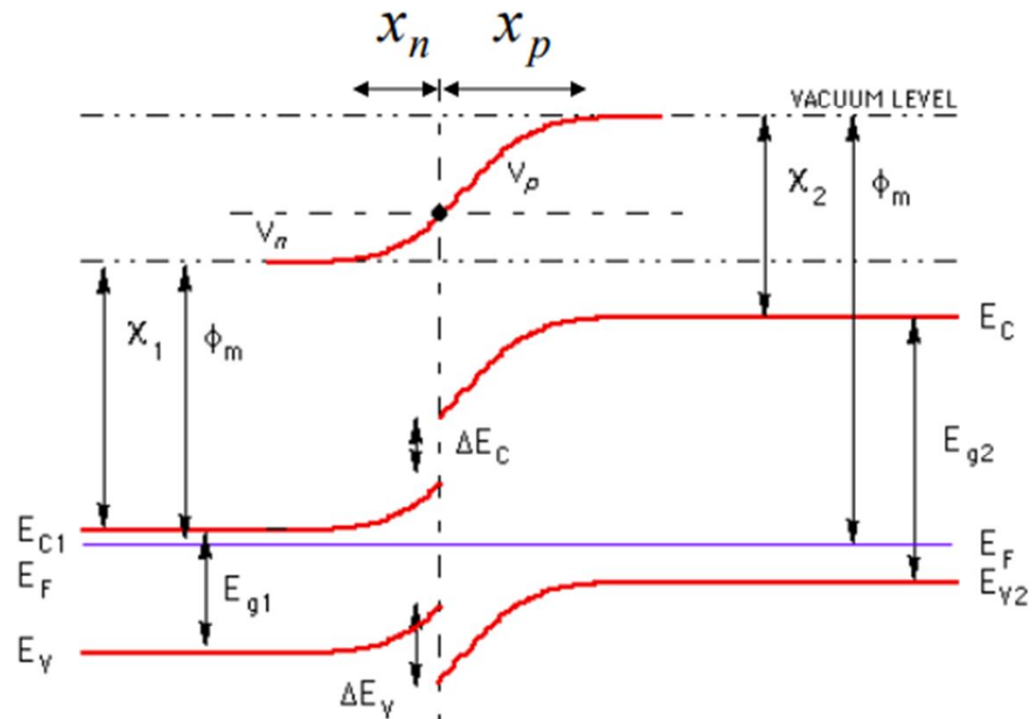
$$x_p = \sqrt{\frac{2\epsilon_p}{q} \frac{\xi}{N_a} (\phi_i - V_a)}$$

$$V_n = (1-\xi)(\phi_i - V_a) - (1-2\xi)V_t$$

$$V_p = \xi(\phi_i - V_a) + (1-2\xi)V_t$$

with
$$\xi = \frac{\epsilon_n N_d}{\epsilon_n N_d + \epsilon_p N_a}$$

Band diagram of a n-P heterojunctions



Reading Materials

D. A. Neamen, “Semiconductor Physics and Devices: Basic Principles”, Fourth edition, Chapter 9, “Metal–Semiconductor and Semiconductor Heterojunctions”.